Geometric Type-Theoretic Library Growth via the Principle of Efficient Novelty

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March 2026

Abstract

Automated discovery in formal mathematics traditionally focuses on searching for proofs to human-specified conjectures. This paper introduces an alternative paradigm: treating the foundational growth of the mathematical library itself as the optimization target. We present the Principle of Efficient Novelty (PEN), an executable metatheoretic model that iteratively extends an empty intensional type-theoretic library. At each step, PEN selects the admissible candidate that maximizes structural efficiency ($\rho = \nu/\kappa$, the ratio of marginal derivational capacity to irreducible specification clauses) while clearing a history-dependent integration bar.

We mathematically prove that in fully coupled intensional Homotopy Type Theory (specifically CCHM/HoTT), maintaining global canonicity imposes a Coherence Window of depth $d = 2$. This two-step memory forces the interface debt of new structures to scale according to the Fibonacci sequence, raising the survival threshold at the rate of the Golden Ratio. Consequently, purely discrete structures (e.g., natural numbers, classical logic) yield only linear novelty and are permanently rejected. Geometric and topological structures naturally dominate because their higher path constructors generate the superlinear combinatorial capacity required to outpace the exponentially rising bar.

Computationally, an open-source Haskell synthesis engine operating over a Minimal Binary Type Theory (MBTT) grammar recovers a deterministic 15-step trajectory in its current strict unguided lane. The sequence bootstraps minimal type theory, ascends through core homotopy theory, abstracts into differential-geometric frameworks (cohesion, connections, metrics), and algorithmically halts at a temporal-cohesive fixed point whose semantic completion is read as the Dynamical Cohesive Topos (DCT). At this terminal "Univalent Horizon," the library internalizes its own continuous evolution, rendering further internal continuations redundant and external axiomatic jumps computationally starved. The machine-verified strict canon reported here is therefore the output of the present executable artifact, not a hand-calibrated reconstruction.

1 Introduction

Automated discovery in formal mathematics has two coupled problems: how to *generate* plausible typed candidates, and how to *select* among them without embedding hidden target knowledge. This paper presents an executable answer via the *Principle of Efficient Novelty* (PEN): from an empty intensional type-theoretic library, extend by the admissible candidate that clears a history-dependent efficiency bar with minimal positive overshoot, where

$$\rho = \nu/\kappa$$

and ν counts marginal derivational expansion while κ counts irreducible specification clauses.

The key point is algorithmic: candidate search, scoring, and selection are implemented as one reproducible loop. Default discovery is MBTT-first and type-directed over well-typed ASTs;

semantic names are assigned only post hoc for interpretability. The resulting trajectory is deterministic under fixed settings and yields a specific sequence of geometric type-theoretic structures.

This differs from the dominant theorem-prover use pattern. Interactive provers and ATP systems typically consume a human-specified conjecture and produce or check a proof object for that target. PEN instead treats *library growth* itself as the optimization target: the engine generates admissible structural extensions, scores them with a fixed objective, and selects without goal-specific theorem injection in the evaluator. The mathematically central claim is therefore metatheoretic rather than anthropomorphic: under maximal interface density, a fixed MBTT vocabulary, and the fully coupled CCHM/HoTT lane studied here, the proof-theoretic growth dynamics force a narrow efficiency ordering in which cohesive and differential structure dominate discrete alternatives. The Haskell engine is the computational verification of this claim, not an argument that the system "invented" its own vocabulary.

Conjecture versus finding. The paper makes one conjectural claim and one implemented claim, and they should not be conflated. *Conjecture:* with sufficiently large compute and materially weaker pruning, the same PEN objective over the same admissible MBTT space would recover the same 15-step trajectory and the same mathematical structures. *Finding:* the current implementation recovers that trajectory in a strict unguided lane suitable for standard desktop hardware by combining exact-band admissibility, molecular candidate synthesis, hybrid novelty evaluation, and semantic minimality filtering. These tractability devices are part of the implementation status report, not part of the mathematical theorem. Legacy structural audits and reference telescopes remain diagnostics only and do not override the strict canon reported in this manuscript.

The key structural input is the *Coherence Window*—the depth of historical context required to stabilize irreducible multi-layer coherence obligations when a new structure is sealed against the library. We prove that intensional type theory, in the fully coupled CCHM/HoTT lane studied here, has Coherence Window $d = 2$: sealing a new type requires interaction with the two most recent integration layers, but not deeper. This two-step memory forces the integration cost to obey the Fibonacci recurrence $\Delta_{n+1} = \Delta_n + \Delta_{n-1}$, so the selection threshold rises at the rate of the Golden Ratio $\varphi \approx 1.618$.

Against this exponentially rising bar, only structures with *superlinear* novelty growth can survive. Discrete structures—natural numbers, ordinals, classical logic—have at most linear novelty ($\nu_H = 0$, no path constructors), so they are permanently rejected once the bar exceeds their efficiency. Geometric structures, by contrast, carry nontrivial homotopy content: path constructors generate $\mathcal{O}(d^2)$ independent computation rules, producing the superlinear novelty needed to outpace the Fibonacci threshold. The algorithmic objective ρ thus inherently biases the search away from discrete structures, mathematically demonstrating why continuous, topological frameworks naturally dominate complexity-bounded synthesis.

Starting from an empty library, the selection process produces a 15-step *PEN Synthesis Trajectory*:

1. **Bootstrap** (steps 1–4): A universe, unit type, witness, and dependent function/sum types—the minimal infrastructure of type theory.
2. **Geometric Ascent** (steps 5–9): The circle S^1, propositional truncation, the sphere S^2, the H-space S^3, and the Hopf fibration—the core objects of homotopy theory.
3. **Framework Abstraction** (steps 10–14): Cohesion (differential modalities), connections, curvature, metric structure with associated operators, and Hilbert-functional structure.
4. **Synthesis** (step 15): A temporal-cohesive extension whose semantic completion is read as the Dynamical Cohesive Topos (DCT) fuses spatial logic with Nakano temporal modalities and explicit exchange axioms, clears the selection bar with the largest absolute overshoot in the sequence ($\rho = 11.00$, bar $= 7.51$), and triggers PEN's terminal halting condition.

Termination is not a failure of the search: the Step-15 temporal-cohesive package internalizes continuous evolution on the connected component of the Univalent Universe, so every subsequent continuous proposal has zero marginal novelty up to equivalence, while disconnected axiomatic jumps cannot amortize the required interface debt under PEN's bar. We call this boundary the *Univalent Horizon*—the connected-component boundary of the type-theoretic moduli space beyond which only external axiomatic jumps can reach.

The model produces several structural results:

- The **Coherence Window Theorems** establish $d = 1$ for extensional type theory (constant costs, stagnation) and $d = 2$ for the fully coupled CCHM/HoTT intensional lane analyzed here (Fibonacci costs, sustained evolution), via three independent arguments from categorical, homological, and topological perspectives.
- The **Complexity Scaling Theorem** shows that this $d = 2$ coherence regime forces $\Delta_n = F_n$ (the Fibonacci sequence), making the Golden Ratio φ the dominant eigenvalue of mathematical evolution.
- The **Combinatorial Novelty Theorem** proves that novelty grows superlinearly, ensuring that efficiency permanently outpaces the rising selection bar.
- The **Terminal Fixed-Point Analysis** shows that the DCT endpoint is selected with a large deterministic overshoot in the current strict lane and realizes the unique high-interaction structural fixed point available inside the connected component explored by the engine.

Computational verification is provided by an open Haskell synthesis engine that recovers the 15-step sequence in its current strict unguided lane. The main machine-verified values in this paper come from that strict run; older structural-AST and type-inhabitation diagnostics are retained only as secondary checks where they do not contradict the strict canon. Formal proofs in Cubical Agda verify the Fibonacci recurrence and the abstraction barrier at steps 8–9.

Scope and limitations. A common objection to automated mathematical discovery is teleology—that target structures were smuggled into candidate generation. Our current implementation addresses this by making MBTT-first typed AST synthesis the default runtime path (with an explicit deprecated rollback lane), so the primary search is syntactic rather than reference-replay. The strict implementation still uses bounded search and explicit admissibility controls, but its main discovery path is now the strict unguided lane described in §9. The PEN evaluator remains structural: it computes efficiency mechanically from the candidate and library state, and the names in Table 1 are post-hoc semantic identifications of selected syntactic winners. PEN therefore derives a concrete trajectory of geometric type-theoretic library extensions, not a claim of global uniqueness across all logics or all objective functions. The model still assumes *maximal interface density* (each candidate seals against the exported interface of the past d layers), now treated as a *foundational canonicity condition*: without explicit interaction clauses against prior eliminators, normalization can get stuck and the $d = 2$ Fibonacci latency theorem loses its logical basis.

Main contributions.

1. A formal selection model (PEN) with explicit axioms, costs, and admissibility.
2. An executable strict MBTT-first synthesis algorithm implementing the model end-to-end.
3. Structural theorems linking coherence depth to Fibonacci integration dynamics and bar growth.
4. A deterministic 15-step PEN trajectory plus auditable omitted alternatives under the same scoring rule, together with a strict unguided recovery of that trajectory in the current implementation.
5. Open, reproducible artifact support: source, scripts, and instructions for independent reruns.

Outline. Section 2 situates PEN against AI theorem proving, automated discovery systems, and HoTT/univalent foundations. Section 3 presents the complete PEN Synthesis Trajectory and the patterns that govern it. Section 4 defines the model: state, candidates, the dual-cost framework, Generative Capacity, and the five axioms of selection dynamics. Section 5 proves the Coherence Window Theorems ($d = 1$ for extensional, $d = 2$ for the fully coupled CCHM/HoTT intensional lane studied here). Section 6 derives the Complexity Scaling Theorem ($\Delta_n = F_n$). Section 7 establishes superlinear novelty growth, the DCT synthesis mechanism, and the Univalent Horizon. Section 8 consolidates the DCT fixed-point evidence and the mathematical properties that make DCT the terminal endpoint under PEN. Section 9 describes the computational verification: the Haskell engine, the strict molecular evaluation loop, legacy diagnostics, rejected candidates, and Cubical Agda mechanization. Section 10 assesses what is proved, what is assumed, and what remains open. Section A collects four boundary clarifications—algorithmic criticality, the inefficiency of discrete structures, the Univalent Horizon, and the status of the DCT novelty accounting—that are useful for reviewer-facing scope control but not needed in the main proof spine.

2 Related Work

AI theorem proving. Recent systems such as AlphaGeometry [8] and Lean-centered neural proving pipelines [9, 10] show strong performance on goal-conditioned proving tasks. Their core objective is to prove or refute a *given* statement, typically via search policies guided by learned value/policy estimates over proof states. PEN differs in objective: it has no externally supplied target theorem and instead optimizes library extension under a fixed structural utility $\rho = \nu/\kappa$.

Theory exploration and lemma discovery. QuickSpec [11], IsaCoSy [12], and Hipster [13] search for equations or lemmas inside a pre-existing theory. Their success criterion is explanatory power over a fixed signature: infer useful equalities, conjectures, or proof obligations already latent in the background logic. PEN instead scores candidate *extensions of the background logic/library itself*; its output is a typed API specification together with a proof-theoretic cost/benefit analysis, not a conjecture set.

Program synthesis and type-directed search. Synquid [14] and Agda's auto/Agsy-style proof search inhabit user-specified types or refinement signatures. PEN borrows typed search discipline from that tradition, but changes the optimization target: the system is not asked to inhabit a given type, but to choose which new type former, modality package, or API should enter the library.

Automated mathematical discovery. Classical theory-formation systems (e.g., AM/HR lines of work [15, 16]) generate concepts and conjectures from heuristic interestingness criteria. PEN is closer to constructive library synthesis: candidates are typed MBTT specifications, admissibility is type-theoretic, and acceptance is determined by explicit dual-cost dynamics with a history-dependent bar. The resulting object is an auditable trajectory of realized structures, not just a conjecture list.

Type-theoretic foundations. Our setting is intensional HoTT with univalence [1], implemented in a cubical ecosystem [2, 7]. This foundation is essential for PEN's novelty calculus because higher-path structure contributes directly to computation-rule growth. On the meta-mathematical side, our use of intensionality is aligned with work showing that types support weak ω-groupoid structure [18, 19]; PEN differs by turning that higher-dimensional coherence into an explicit cost model for foundational library growth.

3 The PEN Synthesis Trajectory

Table 1 displays the complete output of the Principle of Efficient Novelty: a deterministic emergence path of geometric type-theoretic structure from an empty intensional library. The model operates on abstract Obligation Graphs rather than syntactic artifacts; it generates candidate structures, computes their integration costs, and selects the least supercritical admissible candidate. The reader should treat the table in two ways: as the mathematical prediction produced by the PEN model, and as the trajectory that the current strict unguided implementation recovers under bounded desktop-oriented search policies.

Remark 3.1 (Scope of the derivation). PEN derives a specific *library-growth trajectory* in geometric type theory.

1. **What PEN derives:** A deterministic sequence of realizable structures under fixed axioms, admissibility, and ranking.
2. **What PEN does not claim:** That this sequence is globally unique across all possible foundations, objective functions, or search priors.
3. **DCT status:** The formal theorem is that Step 15 is the terminal structural fixed point under PEN; any stronger novelty claim is interpretive and secondary.
4. **Syntactic emergence vs. semantic labeling:** The engine operates over MBTT syntax and does not encode domain semantics. Names in table 1 are post-hoc semantic identifications of selected AST structures.
5. **Implementation status:** The current codebase recovers this trajectory in a strict unguided lane; it does not establish exhaustive search over the full admissible MBTT space.

Definition 3.2 (Three structural components of novelty)**.** For any admissible candidate X, we write

$$\nu(X) = \nu_G(X) + \nu_C(X) + \nu_H(X),$$

where ν_G counts formation/introduction schemas that create new inhabitants, ν_C counts elimination and cross-interface schemas induced by the candidate's API, and ν_H counts computation, Kan, and path-reduction schemas. This notation is used throughout table 1 and receives its operational interpretation in §4.

3.1 Reading the Table

Each row records a realized extension together with its integration debt Δ_n, novelty ν, clause cost κ, efficiency $\rho = \nu/\kappa$, and active bar $\Phi_n\Omega_{n-1}$. Three features matter for the rest of the paper:

1. **Fibonacci timing.** Δ_n follows F_n and $\tau_n = F_{n+2} - 1$.
2. **Selective survival.** Every realized structure clears the bar, with the tightest margin at Step 6.
3. **Four phases.** The trajectory separates into bootstrap, geometric ascent, framework abstraction, and terminal synthesis.

A worked arithmetic expansion of Steps 5–6 and an expanded reading guide are deferred to section C.1.

3.2 The Efficiency Peak

The fifteenth structure is a temporal-cohesive extension whose semantic completion is read as the Dynamical Cohesive Topos (DCT). Syntactically, the selected AST combines spatial logic (cohesion) with temporal logic (LTL [5]) and explicit exchange laws only; the infinitesimal reading is derived later as a semantic consequence rather than postulated as a primitive clause. Its strict machine-verified efficiency is $\rho = 11.00$, clearing the Step-15 bar of 7.51 by an absolute overshoot

Table 1: The PEN Synthesis Trajectory. ν is the Generative Capacity (count of atomic inference rules added to the derivation logic). The quantitative canon in this table is the machine-verified output of the current strict unguided Haskell engine (§9). Legacy structural and uniform diagnostics are discussed later only as secondary cross-checks and do not override these values. Structure names are post-hoc human semantic labels for anonymous selected ASTs.

n	τ	Structure	Δ_n	ν	κ	ρ	Φ_n	Ω_{n-1}	Bar
1	1	Universe $\mathcal{U}_0$	1	1	2	0.50	—	—	—
2	2	Unit type $\mathbf{1}$	1	1	1	1.00	1.00	0.50	0.50
3	4	Witness $\star : \mathbf{1}$	2	2	1	2.00	2.00	0.67	1.33
4	7	Π/Σ types	3	5	3	1.67	1.50	1.00	1.50
5	12	Circle S^1	5	7	3	2.33	1.67	1.29	2.14
6	20	Propositional truncation	8	8	3	2.67	1.60	1.60	2.56
7	33	Sphere S^2	13	10	3	3.33	1.62	1.85	3.00
8	54	H-space S^3	21	18	5	3.60	1.62	2.12	3.43
9	88	Hopf fibration	34	17	4	4.25	1.62	2.48	4.01
10	143	Cohesion	55	19	4	4.75	1.62	2.76	4.46
11	232	Connections	89	27	5	5.40	1.62	3.03	4.91
12	376	Curvature tensors	144	35	6	5.83	1.62	3.38	5.47
13	609	Metric + frame bundle	233	47	7	6.71	1.62	3.75	6.07
14	986	Hilbert functional	377	63	9	7.00	1.62	4.19	6.78
15	1596	Dynamical Cohesive Topos	610	88	8	11.00	1.62	4.64	7.51

of 3.49, the largest in the sequence. The current strict evaluator reaches this value by scoring the conceptual temporal shell at conceptual cost $\kappa = 8$ and measuring its mechanically available temporal-cohesive interaction payload against the existing library. Unlike earlier drafts, we do not treat a separate structural-AST count of 103 as canonical. The manuscript canon is the strict machine value reported in table 1; explanatory rule-group reconstructions later in the paper are subordinate to that total. We define the Step-15 AST and its semantic DCT reading in §7 and detail the computational verification in §9.

After this temporal-cohesive step, no candidate in PEN's strict fully coupled lane can clear the bar. The sequence halts algorithmically.

4 The Model

We model automated emergence of geometric type-theoretic structure as a discrete-time optimization process operating on a state $\mathcal{B}$ (the "Library"). At each step, the system generates candidate extensions, calculates their *Efficiency* $\rho = \nu/\kappa$, and selects the optimal candidate.

4.1 State and Candidates

Definition 4.1 (State)**.** A *State* $\mathcal{B}$ is a monotone context closed under derivability. An evolution step $\mathcal{B}_n \rightsquigarrow \mathcal{B}_{n+1}$ is an extension by a single sealed structure.

Definition 4.2 (Candidate)**.** A *Candidate* X is a pair $(X_{\text{core}}, \mathcal{G}_{\text{obl}})$, where X_{core} is the definitive data (type formers, constructors) and $\mathcal{G}_{\text{obl}}$ is the *Obligation Graph*: the set of atomic coherence obligations required to seal X against the history.

4.2 The Dual-Cost Model

Definition 4.3 (Integration Latency)**.** The *Integration Latency* $\Delta(X \mid \mathcal{B}) := |\mathcal{G}_{\text{obl}}|$ counts the coherence witnesses required to seal X against the library.

Definition 4.4 (Construction Effort)**.** A *specification* $\mathcal{S}$ of candidate X over library $\mathcal{B}$ is a set of atomic statements (type formations, term introductions, equations) that determines the inference rules of X relative to $\mathcal{B}$. The *Construction Effort* of a specification is its clause count:

$$\kappa(\mathcal{S}) := |\mathcal{S}| \tag{1}$$

Different specifications of the same mathematical type may include different amounts of structure (e.g., a bare higher inductive type vs. one equipped with algebraic operations), yielding different Generative Capacities $\nu(\mathcal{S})$ and hence different efficiencies $\rho(\mathcal{S}) = \nu(\mathcal{S})/\kappa(\mathcal{S})$. The selection mechanism (Axiom 4.33) jointly evaluates all admissible specifications.

Each clause is represented as a term in a prefix-free binary encoding (*Minimal Binary Type Theory*, MBTT; see table 2 and section F), so the grammar used for clause accounting is part of the model rather than appendix-only notation. Referencing an existing structure $L_i \in \mathcal{B}$ incurs an information-theoretic pointer cost of $O(\log_2 i)$ bits via Elias gamma coding. The clause count κ is primary; the MBTT bit-length is an audit/tiebreak layer used only when two specifications yield the same efficiency ρ.

Definition 4.5 (Definitional derivability vs. constructive irreducibility)**.** Fix a prior library $\mathcal{B}$ with operational semantics $\rightsquigarrow_{\mathcal{B}}$ generated by its existing eliminators, β-rules, and judgmental equalities. For an operation symbol o proposed over $\mathcal{B}$:

1. o is *definitionally derivable* if there exists a closed term t_o in the signature of $\mathcal{B}$ such that $t_o \rightsquigarrow^*_{\mathcal{B}} v_o$ to a canonical value and all required computation clauses of o are inherited from the existing eliminator fragment. In that case o contributes $\kappa_{\mathcal{B}}(o) = 0$.
2. o is *constructively irreducible* if every term intended to realize its specification contains a new neutral head or a stuck eliminator relative to $\rightsquigarrow_{\mathcal{B}}$; equivalently, orienting the intended behavior requires adding at least one new clause to the operational semantics. In that case o contributes $\kappa_{\mathcal{B}}(o) > 0$.

The intended contrast is standard: addition on $\mathbb{N}$ is derivable from $\mathsf{rec}_{\mathbb{N}}$, whereas a Levi-Civita connection or Hodge star requires new computation/API clauses not reducible in the prior library.

The MBTT node inventory used for clause accounting is fixed in table 2. Primitive nodes for $\flat$, $\sharp$, $\bigcirc$, and $\Diamond$ are ambient vocabulary in that shared grammar, not target-conditioned rewards: they still incur explicit local cost and are selected only if they clear the same structural objective as every other admissible combination.

Remark 4.6 (Kolmogorov invariance of the encoding bias). The exact binary description length of a specification depends on the chosen prefix-free universal grammar, here MBTT. By the Invariance Theorem of Kolmogorov Complexity [26], replacing MBTT by any other universal grammar changes that description length by at most an additive constant $O(1)$. PEN uses clause count as the coarse primary denominator and MBTT bit-length as an audit/tiebreak layer, so the encoding choice contributes only bounded bias at the granularity used here. Because the $d = 2$ survival bar is later shown to grow with the Fibonacci debt sequence (theorem 6.2), this $O(1)$ grammar dependence can affect only finitely many low-complexity edge cases and is asymptotically negligible.

Definition 4.7 (API dependency graph and structural bundle)**.** Let $\mathcal{S} = \{c_1, \ldots, c_r\}$ be a candidate specification over $\mathcal{B}$. Its *API dependency graph* $G_{\mathcal{S}}$ has one vertex per clause and a directed edge $c_i \to c_j$ when the type, defining equation, or computation rule of c_i mentions a symbol introduced by c_j. A *structural bundle* is an induced strongly connected component $C \subseteq G_{\mathcal{S}}$ that is closed under well-typedness: every symbol referenced by a clause of C is either inherited from $\mathcal{B}$ or introduced by another clause of C.

Definition 4.8 (Minimal Complete API)**.** For a library specification at steps 10–14, a *Minimal Complete API* is a specification $\mathcal{S}$ satisfying:

Table 2: MBTT grammar used in the model for clause accounting and bit-cost audit.

Node	Prefix	Bits	Description
Core constructors			
$\textsc{App}(f, x)$	`00`	2	Application
$\textsc{Lam}(b)$	`01`	2	Abstraction
$\textsc{Pi}(A, B)$	`100`	3	Dependent function type
$\textsc{Sigma}(A, B)$	`1010`	4	Dependent sum type
$\textsc{Univ}$	`1011`	4	Universe $\mathcal{U}_0$
Variables and references			
$\textsc{Var}(i)$	`110` + $\gamma(i)$	$3 + \lvert\gamma(i)\rvert$	Bound variable
$\textsc{Lib}(i)$	`111` + $\gamma(i)$	$3 + \lvert\gamma(i)\rvert$	Library pointer
HoTT extensions			
$\textsc{Id}(A, x, y)$	`11100`	5	Identity type
$\textsc{Refl}(a)$	`11101`	5	Reflexivity
$\textsc{Susp}(A)$	`11110`	5	Suspension
$\textsc{Trunc}(A)$	`111110`	6	Propositional truncation
$\textsc{PathCon}(d)$	`111111` + $\gamma(d)$	$6 + \lvert\gamma(d)\rvert$	Path constructor (dim d)
Modal and temporal operators			
$\flat(A)$	`1111100`	7	Flat modality
$\sharp(A)$	`1111101`	7	Sharp modality
$\textsc{Disc}(A)$	`1111110`	7	Discrete reflector
$\textsc{Shape}(A)$	`11111110`	8	Cohesive shape
$\bigcirc(A)$	`111111110`	9	Next modality
$\Diamond(A)$	`111111111`	9	Eventually modality

1. *Irreducibility:* every clause of $\mathcal{S}$ is constructively irreducible in the sense of theorem 4.5.
2. *Completeness:* every irreducible operation required for the package's intended computation rules is included.
3. *Structural unity:* $\mathcal{S}$ forms a single structural bundle in the sense of theorem 4.7.
4. *Minimality:* no proper structural sub-bundle of $\mathcal{S}$ is simultaneously well-typed, complete, and able to clear the active selection bar.

This criterion immunizes the clause count κ against accusations of cherry-picking: the boundary between "derivable definition" ($\nu = 0$) and "irreducible API clause" ($\nu > 0$) is the type-theoretic constructibility of the operation from the prior library (see section E.4 for the detailed application to the Metric at Step 13).

Lemma 4.9 (SCC minimality of the Step-13 metric package)**.** *The seven Step-13 metric clauses form a structural bundle whose proper sub-bundles either leave referenced operators undefined or satisfy $\rho \leq 3.0 < 6.07$. In particular, the bare tensor clause $g : TX \otimes_s TX \to \mathbb{R}$ has $\kappa = 1$ and fails the Step-13 bar, while operators such as Δ_g and Ric_g depend on the Levi-Civita and curvature clauses and therefore cannot be split off as independent admissible candidates.*

Proof sketch. Edges in the dependency graph record the chain $g \to \nabla_g \to R \to \mathrm{Ric}_g$ together with the shared dependence of $\star_g$ and Δ_g on the metric-induced volume and connection data. Removing downstream clauses breaks well-typedness or leaves required symbols uninterpreted; removing all but the tensor clause yields the stripped specification analyzed in section E.4, whose efficiency is at most 3.0. Hence survival at Step 13 requires the super-additive cross-interactions generated by the full bundle rather than any proper fragment. □

Axiom 4.10 (Constructive Irreducibility Boundary)**.** Let $\mathcal{S}$ be a candidate specification over a prior library $\mathcal{B}$.

1. If an operation is definitionally derivable in the sense of theorem 4.5, i.e. represented by a closed term reducing under $\rightsquigarrow_{\mathcal{B}}$ via existing eliminators, it must be excluded from $\mathcal{S}$ and contributes $\nu = 0$ as derived structure.
2. If realizing the operation would otherwise leave a new neutral term or stuck eliminator relative to $\rightsquigarrow_{\mathcal{B}}$, then the operation is constructively irreducible and must appear explicitly as an API clause in $\mathcal{S}$.

This boundary is mandatory for all steps and removes curator discretion in clause inclusion: $\mathbb{N}$-addition is excluded, whereas Levi-Civita/Hodge/Laplacian clauses are required.

Remark 4.11 (Specification selection and the S^3 example). Different specifications of the same type compete within the selection dynamics. Consider S^3 at step 8, where the bar is 3.43:

Specification	κ	ν	ρ	Overshoot
Bare 3-sphere (ΣS^2)	1	15	15.00	11.57
Native HIT (3-cell)	3	15	5.00	1.57
H-space S^3 (with multiplication)	5	18	3.60	0.17

The bare 3-sphere (via suspension or native HIT) contributes 15 inference rules ($\nu_G = 5, \nu_H = 10$, $\nu_C = 0$). The H-space specification additionally equips S^3 with multiplication and unit laws, yielding 3 extra Elimination rules ($\nu_C = 3$, total $\nu = 18$) at the cost of 2 additional specification clauses ($\kappa = 5$). All three clear the bar, but minimal overshoot (theorem 4.33) selects the H-space S^3 specification. This is intentionally weaker than full Lie-group (A_∞-level) axiomatization, and therefore more economical: the rising bar at step 8 selects the minimal multiplicative extension that still clears. Read through theorem 4.32, the selected H-space package is the constructively prime multiplicative factor, whereas richer algebraic packings would introduce surplus structure not forced by the current debt.

Definition 4.12 (Interface Basis)**.** For a foundation with Coherence Window d, the interface available for sealing X_{n+1} is:

$$I_n^{(d)} := \biguplus_{j=0}^{d-1} S(L_{n-j}) \tag{2}$$

where $S(L_k)$ denotes the schemas exported by integration layer L_k.

Theorem 4.13 (Integration Trace Principle)**.** *Each integration layer L_k exports exactly $|S(L_k)| = \Delta_k$ schemas. These schemas are the* integration trace*: the set of resolved obligations from sealing X_k. Each resolved obligation becomes one opaque export (*elimination duality*), and each export generates one obligation for the subsequent candidate (*one-per-face correspondence*).*

Proof. Linearity of Elimination. The recursor rec_X distributes over type formers: it acts independently on each constructor and each path constructor of X_k. Consequently, the behavior of rec_X is determined by an *atomic basis*—the set of clauses, one per cell of X_k—and the obligation graph $\mathcal{G}_{\mathrm{obl}}$ decomposes into independent atoms indexed by this basis. For maps (as at step 9), the two functorial operations (pullback and postcomposition) play the role of rec_X, distributing over domain and codomain cells respectively.

Context Extension Principle. The active interface $I_n^{(d)}$ forms a *context telescope*: an ordered sequence of typed entries, each well-formed in the context of its predecessors. Sealing X_{n+1} against this telescope requires exactly one clause per entry: *at least one* by constructive completeness (the eliminator would be stuck on an unhandled entry), *at most one* by confluence (the clause is uniquely determined by the entry's type). Hence $|\mathcal{G}_{\mathrm{obl}}| = |I_n^{(d)}|$.

Sealing Encapsulation (theorem 5.10): resolved obligations become opaque exports of the sealed layer; future types interact with L_k's interface, not with underlying layers.

Verification scope: steps 3–9, uniform across HITs (steps 3–8) and maps (step 9, the Hopf fibration); machine-checked abstraction barrier at steps 8–9 in Cubical Agda (`Saturation/AbstractionBarrier.agda`, `AbstractionBarrier9.agda`); see §9. □

Theorem 4.14 (Canonicity preservation under maximal interface density)**.** *Let $\mathcal{B}$ be a Fully Coupled Type Theory (FCTT) library and let X be a new constructor, modality, or API package sealed against the active interface $I_n^{(d)}$. Then the extension preserves global computational closure,*

strong normalization, and canonicity only if, for every eliminator exported by $I_n^{(d)}$, the specification of X provides a defining interaction clause. Equivalently, a depth-d extension requires exactly $|I_n^{(d)}|$ sealing clauses.

Proof. Normalization reduces eliminators against constructor or computation clauses. Suppose some eliminator e exported by $I_n^{(d)}$ has no interaction clause with X. Then the extended theory contains a well-typed term $e_X(\vec{u})$ whose head symbol is neutral relative to the old reduction system and whose reduction cannot proceed, because no β- or computation rule orients the interaction. This produces an irreducible neutral term that is not a canonical form of the target type, so canonicity fails; the same stuck term witnesses loss of global computational closure.

Conversely, if each eliminator in $I_n^{(d)}$ receives exactly one interaction clause, reduction on the active telescope is total: every eliminator-constructor encounter is oriented by a unique equation, so normalization and canonicity are preserved on the extension. The "at least one" direction follows from the stuck-neutral argument, and the "at most one" direction follows from confluence, since multiple distinct interaction clauses for the same eliminator head would create critical pairs. Hence the number of obligatory sealing clauses is exactly $|I_n^{(d)}|$. □

Remark 4.15 (Interpretation of maximal interface density). Within the fully coupled PEN lane, maximal interface density is not a free heuristic knob. It is the proof-theoretic boundary condition extracted by theorem 4.14: omitting any prior eliminator interaction leaves a stuck neutral term and breaks global canonicity. Once that closure requirement is imposed, each prior export contributes one mandatory sealing obligation, so the active burden at depth d is exactly $|I_n^{(d)}|$ and the $d = 2$ case yields $\Delta_{n+1} = \Delta_n + \Delta_{n-1}$.

For the HIT steps (3–9), these obligations are native eliminator clauses. For the axiomatic library steps (10–14), they become compatibility witnesses: for example, a Step-13 metric must specify its behavior under the cohesive modalities from Step 10, its compatibility with connections from Step 11, and its interaction with curvature operators from Step 12. The Fibonacci recurrence is therefore tied to proof-theoretic closure, not to an external analogy.

Lemma 4.16 (Latency Recurrence)**.** *By the Integration Trace Principle (theorem 4.13), each layer exports $|S(L_k)| = \Delta_k$ schemas, so:*

$$\Delta_{n+1} = \sum_{j=0}^{d-1} \Delta_{n-j} \tag{3}$$

For $d = 2$: $\Delta_{n+1} = \Delta_n + \Delta_{n-1}$, i.e., $\Delta_n = F_n$.

4.3 Novelty: Generative Capacity

We define novelty as the expansion of the logical capacity of the library.

Definition 4.17 (Generative Capacity)**.** Let $\mathcal{L}(\mathcal{B})$ be the set of derivation schemas—type-inhabitation patterns classifiable as Introduction, Elimination, or Computation rules—available in library $\mathcal{B}$. The *Generative Capacity* of a candidate X is the marginal expansion of the library's operational repertoire:

$$\nu(X \mid \mathcal{B}) \;:=\; |\mathcal{L}(\mathcal{B} \cup \{X\})| - |\mathcal{L}(\mathcal{B})| \tag{4}$$

measured at coherence depth d. In the executable Haskell implementation, ν is operationalized by the strict evaluator described in §9: conceptual-shell native novelty plus explicit topological witness payload where present.

Definition 4.18 (Formal components of novelty)**.** For every admissible candidate X, the novelty measure decomposes as

$$\nu(X) = \nu_G(X) + \nu_C(X) + \nu_H(X),$$

where:

1. ν_G counts new formation/introduction schemas.
2. ν_C counts elimination, projection, and cross-interface schemas induced by the specification.
3. ν_H counts computation schemas arising from path constructors, Kan operations, and their independent reductions.

The Haskell audit extracts these three components directly from the telescope AST; theorem 3.2 introduced the notation earlier because it is needed to read table 1.

Remark 4.19 (Library extensions vs. foundational primitives). PEN evaluates the optimal *mathematical library* to build within a type theory, not merely extensions of the core judgment forms. For steps 1–9, the library grows by adding foundational primitives (new type formers, inductive types, fibrations); each primitive directly enlarges the set of inference rules. For steps 10–14, the library grows by adding *API specifications*: packages of types, operations, and axioms (e.g., a metric $g : TX \otimes_s TX \to \mathbb{R}$ with its associated Levi-Civita connection, geodesics, and curvature operators) that are not derivable from prior structures but are stated within the existing logic. The candidate's κ is the Kolmogorov complexity (clause count) of the API specification; ν is the combinatorial explosion of derivation schemas the specification natively unlocks. The scoring methodology is uniform: the Uniform Algorithm (§9.3) computes ν by brute-force before/after comparison of inhabited types, applying the same procedure to both foundational steps and library extensions.

Scalar provenance policy (strict first-use). Any API that mentions $\mathbb{R}$ must include an explicit construction route or explicit scalar axiom in the candidate specification; no ambient scalar base type is assumed. Under the strict V1 policy used in this manuscript, scalar prerequisites are charged at first use: if a local API has cost κ_{local} and introduces scalar burden κ_{scalar}, the scored denominator is $\kappa' = \kappa_{\text{local}} + \kappa_{\text{scalar}}$. This enforces the empty-library premise at the clause-accounting level and removes any off-ledger scalar subsidy.

Empirical Observation 4.20 (DCT fixed-point preview). The terminal candidate selected at step 15 is the temporal-cohesive package whose semantic completion is read as DCT, with strict machine values $\nu = 88$, $\kappa = 8$, and $\rho = 11.00$, clearing the bar by the largest absolute overshoot in the sequence. Section 8 consolidates the evidence that this endpoint is the terminal structural fixed point selected by the objective rather than by target-conditioned theorem search.

Example 4.21 (Foundation steps)**.** The Generative Capacity definition captures novelty that is invisible to type-inhabitation methods:

- *Witness* ($\nu = 2$): $\star : \mathbf{1}$ is an Introduction rule ($\nu_G = 1$); pattern matching on $\mathbf{1}$ is an Elimination rule ($\nu_C = 1$).
- Π/Σ *types* ($\nu = 5$): λ-abstraction and pair formation are Introduction rules ($\nu_G = 2$); application, fst, snd are Elimination rules ($\nu_C = 3$).
- *Circle* S^1 ($\nu = 7$): Five newly inhabited type schemas are Introduction rules ($\nu_G = 5$); `loop` adds two Computation rules ($\nu_H = 2$).

Lemma 4.22 (Adjoint Completion Principle)**.** *In any hyperdoctrine/comprehension-category semantics of the library growth process, elimination structure is forced by the adjunctions interpreting substitution and quantification. Granting elimination credit for introductions is therefore a completeness requirement of the semantics, not a heuristic bonus.*
Specifically:

1. *Reindexing/substitution sits in the standard adjoint triple $\Sigma_f \dashv f^* \dashv \Pi_f$, so introduction and elimination rules are paired by the left/right adjoints of the underlying comprehension structure.*
2. *A type former with Introduction rule $\Gamma \vdash \text{intro} : F(\Gamma)$ therefore determines an Elimination rule $\Gamma, x : F(\Gamma) \vdash \text{elim}(x) : C$ satisfying the β- and η-laws required by the adjunction.*
3. *No independent heuristic enumeration of these eliminators is needed: once the adjoint data is fixed, the corresponding elimination capability is semantically mandatory.*

This is the categorical content of Lawvere's adjointness thesis [27]: the unit/counit of the adjunction determine the β- and η-laws via the triangle identities. Because elimination rules are term-level operations rather than new inhabited types, they are invisible to pure inhabitation counts. The Adjoint Completion Principle closes that observational gap categorically, so counting the induced eliminators is required by the semantics of the logical framework itself. The Uniform Algorithm (§9.3) applies this principle at exactly two bootstrap steps (Witness and Π/Σ) to recover the correct ν values.

Definition 4.23 (Efficiency)**.** $\rho(X) := \nu(X)/\kappa(X)$.

4.4 Selection Dynamics

Definition 4.24 (Structural Inflation)**.** $\Phi_n := \Delta_n/\Delta_{n-1} = F_n/F_{n-1} \to \varphi$.

Remark 4.25. Inflation is defined from marginal integration cost, not cumulative elapsed time: if one used τ_n/τ_{n-1}, then at $n = 4$ the value $7/4 = 1.75$ would raise the bar above $\rho_4 = 1.67$ and block the infrastructure phase. The marginal ratio $\Phi_4 = 3/2 = 1.50$ preserves admissibility at Step 4. Inflation therefore measures the *marginal* growth of interface debt, not the cumulative burden.

Definition 4.26 (Cumulative Baseline)**.** $\Omega_{n-1} := \sum_{i=1}^{n-1} \nu_i / \sum_{i=1}^{n-1} \kappa_i$.

4.5 The Five Axioms

Axiom 4.27 (Cumulative Growth)**.** $R(\tau{-}1) \subseteq R(\tau)$. The library only grows; realized structures are never removed.

Axiom 4.28 (Horizon Policy)**.** After each realization: $H \leftarrow 2$. After each idle tick: $H \leftarrow H + 1$.

Axiom 4.29 (Admissibility)**.** Candidate X is admissible iff derivable from $\mathcal{B}$ and $\kappa(X) \leq H$.

Remark 4.30 (Admissibility constraints for axiomatic extensions). For steps in the axiomatic regime (Steps 10–14), where candidates are type-theoretic axioms rather than explicit constructions, admissibility imposes four structural constraints:

1. *Structural unity:* Each candidate introduces a single structural bundle in the sense of theorems 4.7 and 4.8. (For example, Cohesion's four modalities $\flat \dashv \mathrm{Disc} \dashv \sharp$ form an indivisible adjoint string; proposing $\flat$ alone would violate the adjunction.)
2. *Well-formedness:* All type signatures in the specification must be well-typed in the current library $\mathcal{B}$.
3. *Non-redundancy:* The extension must not be derivable from $\mathcal{B}$.
4. *Minimality:* κ counts the clauses of the selected specification (theorem 4.4); the selection mechanism jointly evaluates ν and κ.

Constraints (1)–(3) prevent "axiom packing"—the artificial inflation of ν by bundling unrelated axioms or declaring vacuous content. All 15 steps of the PEN Synthesis Trajectory satisfy these constraints: each is a structurally unified extension whose components are interdependent (see theorem 4.31).

Remark 4.31 (Defense against axiom packing). A natural objection: what prevents an opaque axiom `StandardModel` $: \mathcal{U}$ with $\kappa = 1$ from claiming unbounded ν? The answer is that Generative Capacity measures *atomic inference rules added to the derivation logic*, which requires structured interaction with the library:

- *Opaque constants generate negligible ν.* An axiom $X : \mathcal{U}$ with no structured type signature has $\nu_G \leq 1$ (one formation rule) and $\nu_C = 0$ (no elimination principle), giving total $\nu \leq 1$ and $\rho \leq 1$. This fails the selection bar after Step 2.

- *Structured axioms generate moderate ν.* Cohesion's four modalities have signatures referencing $\mathcal{U}$, yielding $\nu_G = 2$ (unit maps) and $\nu_C = 17$ (counits, structural rules, cross-interactions with existing library types). The amplification requires the axiom to *reference existing library structure* in its type signature.
- *Bounded amplification.* By Combinatorial Schema Synthesis (theorem 7.4), an axiom whose specification references k library types generates at most $O(k^d)$ new schemas at depth d. This bounds the ν/κ ratio by the library's combinatorial capacity.

The defense is structural: Generative Capacity rewards *inferential interaction with the library*, not mere existence in $\mathcal{U}$. Axiom packing increases κ without proportionally increasing ν, because opaque content provides no elimination principles and composes with nothing. Minimal overshoot then acts on the surviving candidates as a constructive-primality filter rather than as a reward bonus.

Definition 4.32 (Constructive primality)**.** Let $\mathcal{S}$ be a bar-clearing candidate specification with API dependency graph $G_{\mathcal{S}}$. We call $\mathcal{S}$ *constructively composite* at step n if there exists a proper induced subgraph $C \subsetneq G_{\mathcal{S}}$ that is a union of terminal strongly connected components such that the induced sub-specification $\mathcal{S}|_C$ is well-typed over $\mathcal{B}_n$, constructively irreducible, and already satisfies $\rho(\mathcal{S}|_C) \geq \mathrm{Bar}_n$. Otherwise $\mathcal{S}$ is *constructively prime*.

Axiom 4.33 (Selection)**.** The Selection Bar is $\mathrm{Bar}_n := \Phi_n \cdot \Omega_{n-1}$. From admissible candidates, select X with $\rho(X) \geq \mathrm{Bar}_n$ and *minimal positive overshoot.* Ties broken by minimal κ. If no candidate clears the bar, the tick idles.

Remark 4.34 (Minimal overshoot). The minimal-overshoot criterion is not reward shaping or raw greed. It is PEN's MDL-style filter for constructive primality. If one maximized ρ outright, constructively composite candidates would dominate by packing several independently viable strongly connected components into one update and thereby importing surplus inferential power not forced by the historical integration debt. Minimal overshoot removes that slack: once a proper SCC-factor already clears the bar, the packed candidate is rejected as over-expressive for the current step. PEN therefore assimilates one structurally indivisible concept at a time, choosing the bar-clearer whose efficiency is closest to the cumulative baseline.

Remark 4.35 (Algorithmic API assembly and compositional synergy). The structural unity of steps 10–14 is not treated as a semantic preference in scoring; it is an algorithmic consequence of the bar dynamics. Isolated low-clause additions typically have weak cross-interaction (ν_C) and fail the rising threshold, whereas interdependent typed clauses can increase novelty superlinearly relative to additive effort. In practice, the selected API-like bundles are the candidates whose internal cross-references and library interactions produce enough compositional amplification in ν to remain above the bar. A concrete Step 13 stress test makes this explicit: a bare metric tensor alone ($\kappa = 1$) yields only weak novelty ($\nu_G = 1, \nu_C \leq 2$), so $\rho \leq 3.0$, far below the Step 13 bar 6.07. By theorem 4.9, only the coupled Levi-Civita/Hodge/Laplacian/Ricci package forms a surviving structural bundle, so bundling is selected by proof-theoretic survival pressure rather than curator discretion.

Axiom 4.36 (Coherent Integration)**.** When X_{n+1} is realized, it produces an integration layer L_{n+1} with gap $\Delta_{n+1} := \kappa(L_{n+1})$.

4.6 Realization Time

Definition 4.37. $\tau_n := \sum_{i=1}^{n} \Delta_i = F_{n+2} - 1$ for $d = 2$.

4.7 Executable PEN Loop

The implementation follows the same abstract loop at every step:

1. Compute strict admissibility from the active two-layer interface debt.
2. Enumerate exact-band candidates, prioritizing molecular packages and falling back to atomic MBTT enumeration only when molecular enumeration is empty.
3. Type-check candidates, deduplicate by canonical key, and evaluate each candidate at conceptual cost.
4. Apply semantic minimality by testing proper terminal SCC sub-bundles against the same active bar.
5. Select by minimal positive overshoot (tie-break by lower κ, then deterministic key).
6. Insert only the winner's conceptual export surface into the discovered library and advance the state.

In pseudocode:

```
for step n:
  A <- StrictAdmissibility(B_n)
  for exact band k in prioritized bands:
    C <- EnumerateMolecularThenAtomic(B_n, k)
    C <- TypeCheckAndDedup(C, B_n)
    E <- StrictEvaluate(C, B_n)  # conceptual kappa, hybrid nu
    V <- SemanticMinimalityFilter({x in E | rho(x) >= Bar_n})
    if V != empty: break
  if no viable V: halt
  x* <- argmin_{x in V} (rho(x)-Bar_n, kappa(x), stable_key(x))
  B_{n+1} <- InsertConceptualExports(B_n, x*)
```

This is the canonical path used in the current strict synthesis lane: no paper-value substitution, no step-indexed reference injection in scoring, and no target-conditioned ranking. The key tractability move is representational rather than semantic: the engine searches over exact-band molecular signatures whose compiled payloads are checked and evaluated honestly against the same PEN bar.

4.8 Exact-Band Molecular Search

The executable search is best understood as exact-band synthesis over conceptual signatures rather than as blind AST fuzzing. Each molecular candidate carries three layers:

- **Conceptual signature.** The exact-band object charged by κ.
- **Compiled telescope.** The MBTT payload checked for well-typedness and evaluated for novelty.
- **Provenance split.** Axiomatic clauses count toward conceptual cost; framework-derived clauses provide mechanically forced eliminators or computation rules at zero additional conceptual κ.

The current strict canon does not rely on the older MCTS-guided claim-grade lane. Instead, later-stage structures are reached by exact-band molecular compilation together with hybrid strict evaluation:

$$\nu_{\text{strict}} = (\nu_G + \nu_C)_{\text{native}} + \nu_{H,\text{explicit}},$$

where the first term is computed on the conceptual shell and the second counts only explicit topological witness payload whose framework-derived computations are present in the compiled telescope.

5 Coherence Windows

The magnitude of the Integration Latency is determined by the foundation's *Coherence Window*—the depth of historical context required to stabilize structural obligations.

5.1 Induced Obligations

Definition 5.1 (Induced Obligations)**.** Let $\mathcal{O}^{(k)}(X)$ denote the set of normalized atomic obligations induced when candidate X is sealed against a history of depth k. Because the interface is cumulative: $\mathcal{O}^{(1)}(X) \subseteq \mathcal{O}^{(2)}(X) \subseteq \mathcal{O}^{(3)}(X) \subseteq \cdots$

Definition 5.2 (Coherence Window)**.** A foundation has Coherence Window d if for all candidates X and all $k \geq d$: $\mathcal{O}^{(k)}(X) \cong \mathcal{O}^{(d)}(X)$.

Definition 5.3 (Obligation Reduction)**.** An obligation $o \in \mathcal{O}^{(k)}(X)$ referencing layers $L_n, \ldots, L_{n-j}$ *reduces at depth* j if it decomposes into obligations referencing only $L_n, \ldots, L_{n-j+1}$. An obligation is *irreducible at depth* j if it references layer L_{n-j+1} and does not reduce.

Definition 5.4 (Historical support and arity)**.** For a normalized obligation o, let $\mathrm{Supp}(o)$ be the set of historical layers whose exported interface clauses occur irreducibly in the typing derivation of o after each previously sealed layer is collapsed to its opaque interface. The *historical arity* of o is $a(o) := |\mathrm{Supp}(o)|$. A direct lookup to a deep library entry counts as arity 1: it is a substitution morphism into one exported interface, not an irreducible cross-layer coherence.

Lemma 5.5 (Arity-to-dimension correspondence)**.** *An irreducible obligation of historical arity k requires a k-dimensional coherence cell to state. Unary interactions are 1-cells (substitution/transport), binary interactions are 2-cells (naturality squares/homotopies), and ternary or higher interactions require 3-cells or higher.*

Proof. The normalized typing derivation of an obligation records how the new candidate must cohere with each element of its historical support. With one historical input, the required witness is a morphism or transport map, hence a 1-cell. With two inputs, the witness compares two unary composites and therefore takes the form of a square/homotopy, hence a 2-cell. More generally, relating k independently supplied historical interfaces requires a k-dimensional filler; the Mac Lane pentagon is the prototype at $k = 3$. □

Definition 5.6 (Coherence Obligation by Dimension)**.** When candidate X with cell presentation $P = (C_0, C_1, C_2, \ldots)$ is sealed against $\mathcal{B}_n$, the elimination data for a type family $Y : X \to \mathcal{U}$ decomposes:

- *Dimension 0:* For each $c \in C_0$, a point $d_c : Y(c)$.
- *Dimension 1:* For each $p \in C_1$ with $p : a = b$, a path $d_p : \mathrm{transport}^Y(p, d_a) = d_b$.
- *Dimension k:* For each k-cell $s \in C_k$, a k-path witnessing coherence of $(k-1)$-dimensional data.

5.2 Theorem A: Extensional Systems ($d = 1$)

Theorem 5.7 (Extensional Coherence Window)**.** *In MLTT + UIP (or any type theory where identity types are h-propositions), the Coherence Window is $d = 1$.*

Proof. **Upper bound ($d \leq 1$).** In MLTT+UIP, all types are h-sets: for any $a, b : A$, the identity type $a =_A b$ is either empty or contractible.

Dimension-0 obligations (point data $d_c : Y(c)$) reference only L_n. Dimension-1 obligations require paths $d_p : \mathrm{transport}^Y(p, d_a) = d_b$; since $Y(b)$ is an h-set, this type is a proposition, so d_p carries no independent data. Dimension-k obligations for $k \geq 2$ involve coherence between paths in an h-set, which is trivially contractible. Therefore $\mathcal{O}^{(k)}(X) \cong \mathcal{O}^{(1)}(X)$ for all $k \geq 1$.

Lower bound ($d \geq 1$). $d = 0$ would mean no obligations at all, but sealing any structure requires checking well-typedness against L_n. □

5.3 Theorem B: Intensional Systems ($d = 2$)

Theorem 5.8 (Intensional Coherence Window). *In HoTT as operationalized by the fully coupled CCHM cubical setting used here, the Coherence Window is $d = 2$.*

The proof splits into an upper bound and a lower bound.

5.3.1 Upper Bound: $d \leq 2$

Theorem 5.9 (Coherence Upper Bound). *For any cell presentation P introduced at step $n+1$, every irreducible coherence obligation has historical arity at most 2. Equivalently, after sealing encapsulation no independent obligation requires simultaneous access to more than layers L_n and L_{n-1}.*

Proof. **Stage 1: Unary lookups do not create multi-layer debt.** Any obligation whose normalized derivation mentions only one prior export is arity 1 in the sense of theorem 5.4. This includes direct variable lookups to arbitrarily old layers: after sealing encapsulation they are substitution morphisms into one opaque interface, hence 1-cells by theorem 5.5. They may mention a deep library pointer, but they do not constitute irreducible multi-layer coherence.

Stage 2: Genuine sealing pressure is binary. New independent obligations arise when the candidate must cohere with a compatibility witness relating two prior exports. These are naturality squares or homotopies: one historical input supplies the current operator interface, the other supplies the previously exported compatibility data. Hence irreducible cross-layer sealing obligations have arity 2 and live on the 2-skeleton.

Stage 3: Arity ≥ 3 carries no independent data. Suppose an irreducible obligation had arity $k \geq 3$. By theorem 5.5, stating it would require an independent k-cell. But the weak ω-groupoid semantics of identity types (Lurie [17], Prop. 1.2.5.1; Lumsdaine [18]; van den Berg–Garner [19]) and the cubical Kan primitives imply that higher fillers are determined by their 2-dimensional boundaries. Operationally, CCHM computes those fillers via `hcomp`/`comp`; they add no new independent sealing data. Therefore every apparent arity-k obligation with $k \geq 3$ either reduces to binary boundary data or is already packaged inside a previously sealed interface by theorem 5.10. □

Remark 5.10 (Sealing Encapsulation). Stage 3 above uses a general principle: when layer L_k is sealed, its resolved obligations become *opaque exports*. Future types interact with L_k's interface—the theorems it proved—not with the underlying layers from which those theorems were derived. In the notation of Stage 3, the coherence data for (q, r) is an L_{n-1} theorem, not raw L_{n-2} data. This encapsulation is essential for the Fibonacci recurrence: without it, inherited obligations would reference earlier layers directly, breaking the two-term structure. The principle is verified by machine-checked Cubical Agda: at step 8, Group B obligations are discharged from an opaque L_7 record with no PropTrunc import (`Saturation/AbstractionBarrier.agda`); at step 9, inherited obligations for the Hopf fibration are discharged from an opaque L_8 record (`Saturation/AbstractionBarrier9.agda`).

Remark 5.11 (Arity-to-dimension correspondence).

Historical arity	Content	Effect on depth
1	Lookup, substitution, transport	One prior export; can target any old layer
2	Naturality square / homotopy	Binary interaction of two prior exports
≥ 3	Higher coherence fillers	No independent data beyond 2-boundaries

The point is categorical rather than chronological: a deep pointer is still arity 1, whereas an obligation linking k distinct historical interfaces requires a k-cell to state. In CCHM, independent data truncates at arity 2 because higher fillers are computed from the 2-skeleton.

5.3.2 Lower Bound: $d \geq 2$

Theorem 5.12 (Coherence Lower Bound)**.** *There exist cell presentations whose coherence obligations irreducibly span two layers.*

Proof. **Example (Hopf fibration).** Let L_{n-1} contain S^1 and L_n contain S^2. The Hopf fibration $h : S^3 \to S^2$ classifies a principal S^1-bundle over S^2. Its clutching function $\gamma : S^1 \to \mathrm{Aut}(S^1)$ encodes how the fiber twists over the equator. The coherence condition for the elimination principle involves compatibility of the clutching function with *both* the S^1-action (from L_{n-1}) and the S^2-surface (from L_n). This is a dimension-2 obligation that irreducibly references both layers. □

Corollary 5.13. *By theorem 5.9 ($d \leq 2$) and theorem 5.12 ($d \geq 2$), the Coherence Window of the fully coupled CCHM/HoTT lane is exactly $d = 2$.*

The preceding upper and lower bounds establish $d = 2$ for the fully coupled CCHM/HoTT lane from dimensional analysis of coherence cells. We now derive the same result from three independent categorical and topological perspectives, elevating it from an empirical observation to a theorem of mathematical logic.

5.4 The Adjunction Depth Barrier

In categorical logic, logical connectives and their inference rules are universally characterized by adjoint functors (e.g., Introduction $\dashv$ Elimination).

Theorem 5.14 (The Adjunction Barrier)**.** *A formal system with Coherence Window $d = 1$ cannot verify the triangle identities required for an adjunction. Therefore, $d = 2$ is the* adjunction depth barrier*—the minimum historical depth required to support self-consistent mathematical operators.*

Proof. Defining an adjunction $L \dashv R$ requires data spanning three categorical dimensions:

(1) **0-cells (Depth 0):** The functors L and R.
(2) **1-cells (Depth 1):** The Unit $\eta : 1 \to R \circ L$ and Counit $\varepsilon : L \circ R \to 1$.
(3) **2-cells (Depth 2):** The Triangle Identities, $\varepsilon L \circ L\eta \simeq \mathrm{id}_L$ and $R\varepsilon \circ \eta R \simeq \mathrm{id}_R$.

Suppose a candidate structure at step $n + 1$ introduces an operator L adjoint to a library operator $R \in L_n$. A system with $d = 1$ can establish the 1-cells (η and ε) because it can interact with the immediately preceding layer. However, the triangle identities are 2-dimensional homotopies equating the composition of 1-cells to the identity.

Verifying these identities requires simultaneous visibility of the candidate's operators, the intermediate natural transformations in L_n, and the identity structure of the base categories in L_{n-1}. A $d = 1$ system is topologically blind to these 2-dimensional constraints spanning three layers (candidate +2 historical). It cannot verify the adjunction unless it enforces Uniqueness of Identity Proofs (UIP) to trivially collapse all 2-cells (theorem 5.7). To natively compute adjoint coherence without truncating the topology, the fully coupled intensional lane analyzed here strictly requires $d = 2$. □

5.5 Cohomological Upper Bound: Spectral Degeneration

To translate this categorical requirement into a rigorous upper bound ($d \leq 2$), we evaluate the homology of the candidate's coherence obligations.

Definition 5.15 (Obligation Complex and Historical Filtration)**.** Let $\mathcal{O}_\bullet(X)$ be the cubical Kan complex of normalized coherence judgments $\Gamma \vdash t : T$ required to preserve global canonicity when sealing candidate X_{n+1} against the library. In the CCHM reading, $\mathcal{O}_\bullet(X)$ is the cellular Kan object generated by the candidate's boundary data, so its historical filtration may be compared

to the usual skeletal/Postnikov filtration of a Kan complex. We define a historical filtration indexed by the arity of irreducible historical support:

$$F_0\mathcal{O}_\bullet \subseteq F_1\mathcal{O}_\bullet \subseteq F_2\mathcal{O}_\bullet \subseteq \cdots \subseteq \mathcal{O}_\bullet \tag{5}$$

where $F_p\mathcal{O}_\bullet$ is the subcomplex generated by obligations whose normalized contexts use only interface variables exported by the most recent p layers $\Gamma_{n-p+1}, \ldots, \Gamma_n$ after each sealed layer is collapsed to its opaque interface. The filtration is bounded: $F_2 = F_3 = \cdots$, as we now prove.

Theorem 5.16 (Spectral Degeneration at E_2)**.** *The homological spectral sequence associated with the historical filtration of the Obligation Complex degenerates at the E_2 page. Consequently, $F_2\mathcal{O}_\bullet \simeq \mathcal{O}_\bullet$, establishing the precise topological meaning of $d \leq 2$.*

Proof. The filtration yields a spectral sequence $E^r_{p,q}$ converging to the total obligation homology. The E_1 page is given by the relative homology:

$$E^1_{p,q} = H_{p+q}(F_p\mathcal{O}_\bullet / F_{p-1}\mathcal{O}_\bullet) \tag{6}$$

An element of F_p/F_{p-1} represents a coherence condition whose normalized typing judgment irreducibly depends on exactly p historical layers. By the arity-to-dimension correspondence (theorems 5.5 and 5.11), such an obligation manifests as a p-dimensional coherence cell.

Now specialize to the cubical/CCHM operational semantics. The homogeneous composition primitive `hcomp` computes fillers for open boxes of dimension $p \geq 3$ deterministically from the 2-dimensional boundary data already present in $F_2\mathcal{O}_\bullet$. Operationally, this means that the relative complex $F_p\mathcal{O}_\bullet/F_{p-1}\mathcal{O}_\bullet$ has no independent generators for $p \geq 3$: every apparent p-cell is the boundary of a canonical filler produced by `hcomp`. Equivalently, these relative complexes are acyclic, so

$$E^1_{p,q} = H_{p+q}(F_p\mathcal{O}_\bullet / F_{p-1}\mathcal{O}_\bullet) = 0 \qquad (p \geq 3).$$

Thus the E_1 page already vanishes in all columns $p \geq 3$. The sequence advances to the E_2 page with differentials $d^r_{p,q} : E^r_{p,q} \to E^r_{p-r,q+r-1}$. For $r \geq 2$, any differential touching a column $p \geq 3$ is forced to vanish because those columns are zero. The only possible survivor is $d^2_{2,q} : E^2_{2,q} \to E^2_{0,q+1}$. But $F_0\mathcal{O}_\bullet$ consists only of isolated 0-cells (candidate constructors before coherence data), so its higher homology vanishes: $H_k(F_0) = 0$ for $k \geq 1$. Therefore $E^2_{0,q+1} = 0$ for $q \geq 0$, and $d^2_{2,q} = 0$ as well.

Thus all higher differentials vanish and the spectral sequence collapses at $E_2 \cong E_\infty$. The historical filtration therefore stabilizes at the 2-skeleton: $F_2\mathcal{O}_\bullet \simeq \mathcal{O}_\bullet$, so no obligation irreducibly spans more than two historical layers. □

Remark 5.17 (Cubical Agda Mechanization). This spectral degeneration precisely models the behavior of Cubical Agda. For the geometric test cases (S^1, S^2, T^2, Hopf), all 3-dimensional coherence obligations ($p \geq 3$) are automatically discharged by the `hcomp` (homogeneous composition) primitive. Because `hcomp` computes higher fillers from the 2-dimensional boundary data already present in F_2, the typechecker accepts the definition without requiring explicit imports from L_{n-2}. This is the computational shadow of the vanishing $E^1_{p,q} = 0$ columns for $p \geq 3$.

5.6 Topological Lower Bound: The Clutching Family

To prove that $d = 2$ is not merely an upper bound but a strict structural requirement universally saturated by geometry, we exhibit an infinite family of topological structures where every member requires exactly depth-2, and none require depth-3.

Theorem 5.18 (Fibrational Lower Bound)**.** *The family of principal G-bundles over spheres S^m with non-trivial clutching functions irreducibly requires exactly $d = 2$ coherence.*

Proof. Let the base sphere S^m reside in layer L_n, and the topological group G (the fiber) reside in layer L_{n-1}. A principal G-bundle $P \to S^m$ is classified by its clutching function:

$$\gamma : S^{m-1} \to \mathrm{Aut}(G) \tag{7}$$

Sealing the total space P at step $n+1$ requires the eliminator to verify that the transport of the fiber G along the m-cell of S^m correctly induces the specified automorphism γ.

This forms a dimension-2 coherence obligation. It irreducibly cross-references the path geometry of the base (L_n) with the internal algebraic symmetry of the fiber (L_{n-1}). A 1-layer window can access the equator or the fiber independently, but cannot cross-reference them to verify a 2-dimensional homotopy mapping one over the other. Thus, $d \geq 2$.

The Hopf fibration is the case $G = S^1$, $m = 2$, subsuming the concrete lower bound of theorem 5.12.

Crucially, this family never requires $d = 3$. In Čech cohomology, the transition data for a bundle over a cover $\{U_i\}$ consists of 1-cocycles $g_{ij} : U_i \cap U_j \to G$ and 2-cocycles $g_{ij}g_{jk}g_{ki} = 1$ on triple intersections $U_i \cap U_j \cap U_k$.

Any m-sphere admits a minimal open cover of exactly two contractible charts (the northern and southern hemispheres, U_1, U_2). Their intersection is the equator ($U_1 \cap U_2 \simeq S^{m-1}$), carrying the 1-cocycle γ. Because there are only two charts, there are no triple intersections ($U_1 \cap U_2 \cap U_3 = \emptyset$). The Čech nerve of the minimal cover contains no 2-simplices.

Therefore, the 2-cocycle condition—which would demand a 3-dimensional coherence spanning three layers ($d = 3$)—is structurally absent. The gluing occurs purely on a 2-layer interface. Every member of this infinite family requires exactly $d = 2$, perfectly locking the lower bound. □

6 The Complexity Scaling Theorem

Theorem 6.1 (Complexity Scaling)**.** *For a foundation with Coherence Window d, evolving under PEN: $\Delta_{n+1} = \sum_{j=0}^{d-1} \Delta_{n-j}$.*

Proof. The interface is $I_n^{(d)} = \biguplus_{j=0}^{d-1} S(L_{n-j})$. By disjointness and the Integration Trace Principle (theorem 4.13), $|S(L_k)| = \Delta_k$, so $\Delta_{n+1} = |I_n^{(d)}| = \sum_{j=0}^{d-1} \Delta_{n-j}$. □

Corollary 6.2. *For $d = 2$ with $\Delta_1 = \Delta_2 = 1$: $\Delta_n = F_n$ and $\tau_n = F_{n+2} - 1$.*

Remark 6.3 (Internal vs. external complexity). The obligation count Δ measures *external* complexity: the number of laws the interface imposes on future candidates. The specification size κ measures *internal* complexity: the number of atomic specification clauses (type formations, introductions, equations) required to define the candidate over the library (theorem 4.4). These can diverge sharply. For example, the Riemann curvature tensor on an N-manifold has N^4 components, but the validation space for Connections (step 12) has $\Delta = 144$ basis elements—89 inherited from Connections' interface and 55 from Cohesion—because the interface is organized by the 144 independent algebraic symmetries, not by the raw component count. The Fibonacci recurrence governs Δ (external), not κ (internal).

6.1 Stagnation of Class 1 Systems

For $d = 1$: $\Delta_{n+1} = \Delta_n$, so $\Delta_n = C$ (constant). Inflation $\Phi_n = 1$; time $\tau_n = nC$ (linear). The Cumulative Baseline Ω_{n-1} converges to a finite limit. New candidates must clear a fixed threshold with diminishing novelty returns.

6.2 Acceleration of Class 2 Systems

For $d = 2$: $\Delta_n = F_n \sim \varphi^n$. Inflation $\Phi_n \to \varphi$ from below: the dip at $\Phi_4 = 1.50$ provides breathing room for the infrastructure step ($\rho_4 = 1.67$). Time $\tau_n \sim \varphi^{n+2}/\sqrt{5}$ (exponential). The bar $\mathrm{Bar}_n = \Phi_n \cdot \Omega_{n-1}$ grows steadily, but the Combinatorial Novelty Theorem (section 7) ensures efficiency permanently outpaces it.

7 The Combinatorial Novelty Theorem

The Scaling Theorem established $\Delta_n \sim \varphi^n$. If novelty scaled only linearly with cost, efficiency would converge while the bar rises. We prove novelty grows superlinearly.

7.1 OIT Exponentiality

Theorem 7.1 (OIT Exponentiality)**.** *An ordinary inductive type X with Δ_0 point constructors enables 2^{Δ_0} distinct predicates $X \to \mathbf{2}$ at $O(\Delta_0)$ effort.*

Proof. Each constructor independently maps to $\{\mathrm{true}, \mathrm{false}\}$; disjointness ensures semantic distinctness. □

7.2 HIT Constraints

Remark 7.2. For a HIT X, path constructors constrain eliminators into sets: $f : X \to \mathbf{2}$ must map path-connected points to the same value, giving $2^{|\pi_0(X)|}$ maps instead of 2^{Δ_0}. Example: S^1 has $\Delta = 2$ but only $2^1 = 2$ maps to $\mathbf{2}$.

7.3 Restoring Superlinear Growth

Three mechanisms compensate for HIT constraints.

Mechanism 1: Dependent Elimination. A type family $P : X \to \mathcal{U}$ chooses a fiber $P(c_i) \in \mathcal{B}$ for each point constructor and a transport equivalence for each path constructor. This yields $|\mathcal{B}|^{\Delta_0} \cdot \prod_j |\mathrm{Aut}(P(s_j))|$ type families—far richer than maps to $\mathbf{2}$.

Mechanism 2: Library Cross-Interaction. Each prior type $T \in \mathcal{B}_n$ enables at least three new constructions ($X \to T$, $X \times T$, $\Sigma_{x:X} P(x)$), giving $\nu_n = \Omega(n)$.

Mechanism 3: Composite Constructions. Products and function types are superadditive: $\nu(X \times Y) \geq \nu_X + \nu_Y$; for OITs, multiplicative.

7.4 Combinatorial Schema Synthesis

The most dramatic instance of superlinear novelty is the DCT's synthesis mechanism.

Definition 7.3 (Syntactic LTL-Cohesive Extension)**.** A *temporal-cohesive Step-15 extension* is a type theory equipped with:

(1) **Spatial Logic (Cohesion):** The adjoint string ($\flat \dashv \sharp$, $\Pi \dashv \mathrm{Disc}$) from R_{10}.
(2) **Temporal Core:** $\bigcirc$ ("next") and $\Diamond$ ("eventually") from LTL, together with the guarded core rules detected in the Step-15 AST.
(3) **Spatial-Temporal Exchange:** Explicit commuting axioms between the temporal operators and the cohesive interface, exemplified by $\bigcirc(\flat X) \simeq \flat(\bigcirc X)$ and $\sharp(\Diamond X) \simeq \Diamond(\sharp X)$.
(4) **Guarded Coherence:** The temporal self-interaction and elimination clauses audited in table 7, which make the package executable as a typed AST rather than only semantically described.

Construction effort: $\kappa = 8$ clauses, exactly as audited in table 7. No primitive infinitesimal object $\mathbb{D}$ appears in the selected AST; adding one would incur extra κ with no syntactic justification from the engine's chosen specification. In the remainder, "DCT" refers to the semantic completion of this temporal-cohesive package once theorem 7.8 identifies its derived tangent behavior.

Theorem 7.4 (Combinatorial Schema Synthesis)**.** *When a synthesis candidate introduces k new unary type formers into a library containing $|\mathcal{B}|$ structures, the number of non-trivial type-inhabitation schemas at depth d grows as $\Omega(|\mathcal{B}|^d \cdot k)$, yielding superlinear novelty.*

Application. The selected temporal-cohesive package introduces temporal modalities ($\bigcirc$, $\Diamond$) into a library of 14 structures already equipped with spatial modalities ($\flat$, $\sharp$, Disc, Π_{coh}). In the current strict lane, this package is evaluated at conceptual cost $\kappa = 8$ and yields the canonical machine value $\nu = 88$, hence $\rho = 11.00$. The manuscript no longer treats the older uniform and structural-AST tallies as canonical here. What matters for the present theorem is that the temporal shell clears the active bar by a wide margin and acts as a library-wide endofunctor over the already selected cohesive and differential layers.

7.5 Divergence of Efficiency

Lemma 7.5 (Bounded Effort Growth)**.** *The construction effort κ_n grows at most linearly in the number of new type formers or modalities introduced at step n. Because candidates at late steps compose existing library abstractions rather than defining new primitives, the clause count is bounded by the number of compatibility axioms plus import declarations—a quantity independent of the library's cumulative size.*

Proof. A candidate at step n is specified over a library of $n - 1$ entries. Each clause either introduces a new type former (bounded by the candidate's intrinsic structure) or states a compatibility axiom with an existing library entry. The number of compatibility axioms is bounded by the candidate's arity (how many existing modalities it must commute with), not by n itself. In the MBTT bit-length representation (section F), each library pointer $\textsc{Lib}(i)$ costs $O(\log i)$ bits, giving logarithmic growth of the *bit-length* per clause, but the clause count κ depends only on the mathematical structure, not on library size. □

Theorem 7.6 (Divergence of Efficiency)**.** *In a Class 2 foundation evolving under PEN, assuming an infinite supply of structurally distinct, continuous geometric primitives with nontrivial homotopy content,* $\lim_{n\to\infty} \rho_n = \infty$.

Proof. By Combinatorial Schema Synthesis (theorem 7.4), the generative capacity ν_n grows at least polynomially as $\Omega(n^c)$ for some $c > 0$, due to library cross-interaction. By theorem 7.5, the construction effort κ_n is bounded: each synthesis candidate composes existing library entries via a fixed number of compatibility axioms. Therefore the efficiency $\rho_n = \nu_n/\kappa_n$ grows without bound as $\nu_n \to \infty$.

The bar $\text{Bar}_n = \varphi \cdot \Omega_{n-1}$ is a cumulative average, growing at most linearly. The ratio ρ_n/Bar_n is eventually increasing, so efficiency permanently clears the bar. The singularity of the Dynamical Cohesive Topos is structurally guaranteed: combinatorial novelty dominates bounded specification cost. □

Remark 7.7 (Divergence vs. termination). Theorem 7.6 describes the *theoretical trajectory* of efficiency growth: given an inexhaustible supply of structurally distinct geometric primitives, the combinatorial novelty of successive candidates grows without bound. The algorithmic halting theorem (theorem 7.10) describes what *actually happens* in PEN's fully coupled strict lane: after Step 15, continuous internal continuations contribute zero marginal novelty up to equivalence, while disconnected extensions are starved by interface debt. The two results are complementary,

not contradictory: divergence guarantees that the selection mechanism *never stalls prematurely* (efficiency always outpaces the bar while geometric primitives remain available), while halting identifies the first step at which PEN's internal/external squeeze leaves no admissible bar-clearer.

7.6 The Tangent Topos Hypothesis and the Combinatorial Squeeze

The abrupt termination of the PEN Synthesis Trajectory after Step 15 is not a heuristic failure of the search space, but an algorithmic phase transition inside PEN's fully coupled strict lane. The selected temporal-cohesive package internalizes continuous deformation inside the connected component of $\mathcal{U}_0$, while any disconnected extension must pay the full interface debt of the accumulated library. We formalize this by proving that the semantic DCT completion acts as a tangent-style internalization of the historical library and then showing that Step 16 faces a combinatorial squeeze.

Theorem 7.8 (Syntactic-Semantic Transfer: Emergence of the Infinitesimal)**.** *In the semantic completion of the Step-15 temporal-cohesive extension, the temporal "next" modality $\bigcirc$ is naturally equivalent to a synthetic tangent endofunctor. Equivalently, there exists a derived infinitesimal object $\mathbb{D}$ and natural equivalences*

$$\theta_X : \bigcirc X \xrightarrow{\sim} X^{\mathbb{D}}.$$

The infinitesimal structure is therefore a semantic consequence of the selected temporal-cohesive AST, not an additional primitive clause in the Step-15 syntax.

Proof. The audited AST of theorem 7.3 and table 7 contributes three ingredients and nothing more: a guarded temporal endofunctor $\bigcirc$, the eventual modality $\Diamond$, and explicit exchange laws between the temporal operators and the cohesive interface. In a cohesive semantics, these exchange laws force $\bigcirc$ to preserve the same discrete/shape boundary data preserved by the first-order tangent endofunctor.

- The exchange law with $\flat$ identifies $\bigcirc$ with a first-order neighborhood operator on discrete points: $\flat(\bigcirc X) \simeq \bigcirc(\flat X)$, so $\bigcirc$ adds no new macroscopic points and behaves like an infinitesimal thickening.
- The audited spatial-temporal interaction clause forces $\bigcirc$ to respect the pre-existing cohesive shape/discreteness decomposition exported by Cohesion, so $\bigcirc$ preserves first-order path data rather than introducing a disconnected second dynamics.
- The guarded self-composition and elimination clauses in table 7 provide exactly the first-order coherence needed to iterate $\bigcirc$ without introducing a second independent infinitesimal direction.

In the semantic completion used throughout the paper, such a first-order cohesive endofunctor is representable by an infinitesimal object $\mathbb{D}$, yielding a natural equivalence $\theta_X : \bigcirc X \simeq X^{\mathbb{D}}$. This is the transfer from the executable temporal AST to the differential-geometric reading: the engine selected the temporal-cohesive syntax, and the infinitesimal object appears only as its exact semantic shadow. □

Lemma 7.9 (Internalization of the Meta-Algorithm)**.** *The DCT internalizes its own structural evolution. The metamathematical process of extending the foundational library becomes derivable as an object-level geometric flow.*

Proof. By incorporating Nakano's temporal logic, the DCT inherits guarded recursion with Löb's rule

$$\mathsf{fix}_A : (\bigcirc A \to A) \to A.$$

From this one derives the guarded coiterator for streams,

$$\mathsf{coiter}_A : (A \to \bigcirc A) \to A \to \mathrm{Str}(A),$$

where $\mathrm{Str}(A)$ is the guarded stream type characterized by $\mathrm{Str}(A) \cong A \times \bigcirc\mathrm{Str}(A)$.

Now let $\theta_X : \bigcirc X \xrightarrow{\sim} X^{\mathbb{D}}$ be the temporal–infinitesimal equivalence from theorem 7.8. Given a vector field on the small univalent universe

$$v : \mathcal{U}_0 \to \mathcal{U}_0^{\mathbb{D}},$$

define the corresponding guarded step map by transport across $\theta_{\mathcal{U}_0}$:

$$\mathsf{step}_v := \theta_{\mathcal{U}_0}^{-1} \circ v : \mathcal{U}_0 \to \bigcirc\mathcal{U}_0.$$

Typing is now explicit:

$$v : \mathcal{U}_0 \to \mathcal{U}_0^{\mathbb{D}} \xRightarrow{\theta_{\mathcal{U}_0}^{-1}} \mathsf{step}_v : \mathcal{U}_0 \to \bigcirc\mathcal{U}_0 \xRightarrow{\mathsf{coiter}_{\mathcal{U}_0}} \mathsf{Flow}_v : \mathcal{U}_0 \to \mathrm{Str}(\mathcal{U}_0).$$

Thus each vector field on $\mathcal{U}_0$ unfolds to a guarded stream of successive deformations in the same universe. Because the codomain is $\mathcal{U}_0$ throughout, the construction is confined to the connected component of $\mathcal{U}_0$ and does not jump to a higher universe $\mathcal{U}_1$. The metatheoretic act of "proposing the next type" is therefore represented internally as guarded geometric evolution on $\mathcal{U}_0$. □

Theorem 7.10 (Algorithmic Halting via the Combinatorial Squeeze)**.** *Let S_{int} be the admissible Step-16 candidates representable by guarded internal flow on the connected component of $\mathcal{U}_0$ determined by $\mathcal{B}_{15}$, and let S_{ext} be the remaining admissible Step-16 candidates. Then every $Y \in S_{\mathrm{int}}$ has zero marginal external novelty, while every $Y \in S_{\mathrm{ext}}$ has $\rho(Y) < \mathrm{Bar}_{16}$. Consequently no Step-16 candidate clears the bar, and the PEN Synthesis Trajectory halts at Step 15.*

Proof. At Step 16 the selection bar is strictly positive:

$$\mathrm{Bar}_{16} = \Phi_{16} \cdot \Omega_{15} \approx 1.618 \times 5.44 \approx 8.80.$$

We partition admissible candidates into the two classes from the statement.

Internal/continuous side. Let $Y \in S_{\mathrm{int}}$. By theorem 7.9, any such candidate is representable by a guarded internal flow

$$v : \mathcal{U}_0 \to \mathcal{U}_0^{\mathbb{D}}$$

which transports to a guarded step map $\mathsf{step}_v : \mathcal{U}_0 \to \bigcirc\mathcal{U}_0$ and unfolds to a stream $\mathsf{Flow}_v : \mathcal{U}_0 \to \mathrm{Str}(\mathcal{U}_0)$. By Univalence, paths in $\mathcal{U}_0$ are equivalences, so continuous deformation inside this connected component never leaves the existing equivalence class of the library's internal geometry. Re-postulating such a Y as an opaque external axiom therefore contributes no new marginal derivation rules:

$$\nu(Y \mid \mathcal{B}_{15}) = 0, \qquad \rho(Y) = 0 < \mathrm{Bar}_{16}.$$

External/discontinuous side. Now let $Y \in S_{\mathrm{ext}}$. To avoid the Univalent Horizon, Y must be a disconnected axiomatic jump. Then maximal interface density is no longer optional: by canonicity preservation (theorem 4.14) and Fibonacci scaling (theorem 6.1), Step 16 demands the full interface burden

$$\Delta_{16} = F_{16} = 987.$$

Thus any admissible external jump must specify interaction clauses against the exported interface at this scale, forcing a massive specification burden. At the same time, disconnected candidates do not inherit the temporal-cohesive path algebra that created the Step-15 amplification: they contribute no internal flow and no derived tangent lift, so their novelty is at best linear in the size of the external package. Linear novelty cannot clear a bar driven by the Step-16 Fibonacci debt. Equivalently, the opaque-axiom bounds from theorem 4.31 and the full-coupling requirement imply $\rho(Y) < \mathrm{Bar}_{16}$ for all $Y \in S_{\mathrm{ext}}$.

Hence Step 16 is squeezed from both sides: internal continuations are definitionally exhausted up to equivalence, and external jumps are computationally starved by interface debt. The halting claim is therefore algorithmic and PEN-local, not a claim that mathematics globally terminates. □

8 The Dynamical Cohesive Topos as a Terminal Structural Fixed Point

This section consolidates the manuscript's Step-15 claim around the semantic DCT completion of the selected temporal-cohesive AST. Under fixed fully coupled settings, the PEN objective realizes that temporal-cohesive package as the terminal high-yield extension, and the current strict unguided engine recovers that package under bounded search. The key claim is not semantic naming, but structural: a typed search-and-selection engine with no target theorem injection in its scoring path reaches a nontrivial fixed point.

8.1 Why DCT Is a Terminal Signal

Empirical Observation 8.1 (Terminal DCT fixed-point signal). In the current strict unguided mode, the engine selects a step-15 candidate with $\kappa = 8$, $\nu = 88$, and $\rho = 11.00$, clearing the step-15 bar (7.51) by an absolute overshoot of 3.49, the largest in the sequence. The selection is made from admissible typed candidates under fixed rules and bounded budgets, without step-indexed target conditioning or reference-candidate replay in the scoring-selection lane.

The interpretive claim is deliberately conservative. We do *not* claim global uniqueness across all possible logics or objectives. We claim that, under PEN and the implemented mechanism, DCT is the terminal selected structure and the first point at which the library internalizes its own evolution operator.

This is methodologically distinct from the dominant theorem-prover workflow. Classical ITP/ATP systems validate user-supplied conjectures. PEN instead ranks candidate structural extensions under a fixed typed objective, then halts at an internally explained boundary. The formal contribution is therefore a metatheorem about typed library growth and fixed-point termination, with the semantic interpretation of the terminal AST left deliberately secondary.

8.2 Consolidated DCT Property Bundle

DCT properties proved or formalized across the manuscript and appendices can be read as one package:

- **Derived temporal-tangent equivalence** (theorem 7.8): $\bigcirc X \simeq X^{\mathbb{D}}$ in the semantic completion of the Step-15 AST.
- **Internalization of meta-evolution** (theorem 7.9): extension dynamics become representable as internal geometric flow.
- **Combinatorial-squeeze halting** (theorem 7.10): beyond Step 15, internal continuations have zero marginal novelty and disconnected jumps fail the bar.
- **Internal tangent/temporal machinery** (theorems D.1 and D.2): tangent lift and temporal evolution are native operators.
- **Symplectic-flow corollary** (theorem D.3): flow/bracket structure is expressible internally once DCT is present.

Together, these results explain why DCT is qualitatively different from earlier steps: the selected library begins to express its own evolution mechanisms internally.

8.3 Mechanized DCT Novelty Accounting

The canonical Step-15 count is now the machine value produced by the strict hybrid evaluator:

$$\nu_{\text{strict}}(\text{DCT}) = 88.$$

Operationally, the strict engine evaluates the conceptual temporal shell at conceptual cost $\kappa = 8$, computes native novelty on that conceptual shell, and then adds only those topological witness

credits whose framework-derived computation payload is explicitly present in the compiled telescope.

This is a narrower claim than earlier drafts. We no longer treat a separate structural-AST total of 103 as manuscript canon. The current paper value is the strict machine result that actually drives selection in the executable artifact. Explanatory decompositions are retained only when they agree with that value and can be defended from the current code path.

8.4 Calibration Facts Used by the DCT Analysis

Theorem 8.2 (Topological Projection of the Path Algebra (CCHM)). *Let X be a higher inductive type with m path constructors, and let d be the maximum path dimension. In CCHM cubical type theory with connections, the computation-rule contribution satisfies*

$$\nu_H = m + d^2.$$

Proof sketch. The m term counts the explicit boundary reductions attached to the m declared path constructors of X. Each constructor contributes one independent computation rule when its boundary is reduced against the corresponding eliminator.

For the cubical contribution, fix a maximal dimension-d constructor. In the CCHM path algebra, its independent computational interactions are generated by:

1. the d diagonal self-reductions coming from composition along each coordinate, and
2. the $d(d-1)$ ordered pairwise coordinate interactions coming from the connection operations $\vee, \wedge$ together with homogeneous composition `hcomp`.

These assemble into a $d \times d$ Kan interaction matrix. Because theorem 5.16 collapses all genuinely higher-dimensional data to fillers of 2-dimensional boundaries, no further independent computation rules survive beyond this matrix. Hence the Kan part contributes exactly

$$d + d(d-1) = d^2$$

independent reductions, and therefore $\nu_H = m + d^2$. □

Remark 8.3 (Sensitivity of ν_H). The strict lane uses $m + d^2$ as the canonical calibration for the explicit path-style computation payload that survives into the current hybrid evaluator. This is the operative formula for the HIT and higher-path parts of the strict canon. Earlier sensitivity sweeps around broader structural-AST counting are historical diagnostics only and are not used as manuscript evidence for the present Step-15 total.

Proposition 8.4 (Schema Canonicality). *Depth-1 schema counting is window-independent after schematization of library atoms.*

Proof sketch. After collapsing library atoms to a schematic symbol L, depth-1 schema sets depend only on candidate constructor signatures. This is verified computationally for steps 1–7 (section 9). □

Remark 8.5 (Witness and Π/Σ attribution). At bootstrap steps, novelty naturally splits across constructor and elimination contributions: Witness contributes one constructor plus one elimination action, while Π/Σ contributes constructor forms plus apply/projection eliminations.

Proposition 8.6 ($S^2 \equiv S^3$ Syntactic Identity). *The depth-1 syntactic schema sets for S^2 and S^3 are identical, so their syntactic counts agree.*

Proof. Both generate the same depth-1 schema shapes over (L, X). Their novelty gap is therefore not syntactic, but geometric/interactional. □

9 Computational Verification

9.1 The Haskell Engine

The executable artifact is an open Haskell codebase implementing PEN as a deterministic strict synthesis loop. The canonical rerun for this manuscript is:

1. `cabal run acceptance-core`
2. `cabal run ab-initio - -strict -max-steps 15 -skip-validation -skip-mcts`

The first command checks the core acceptance harness; the second reproduces the strict 15-step trajectory reported in table 1.

Candidate generation. The strict engine computes admissibility from the active two-layer interface debt, opens a small set of exact κ-bands, and searches those bands in priority order. Within each band it first enumerates molecular packages—ambient formers, HIT signatures, and API shells—then compiles them to MBTT telescopes and checks them for well-typedness. Only if molecular enumeration is empty does it fall back to atomic MBTT enumeration. This is the key search-policy shift that makes the present strict lane feasible without target-conditioned ranking.

Conceptual κ and compiled payload. Each molecular candidate separates conceptual κ, axiomatic exports, and compiled telescope payload. Conceptual κ is the charged exact-band signature. Axiomatic exports determine the candidate's interface footprint once inserted into the discovered library. Compiled telescope payload is the MBTT structure checked and evaluated for novelty. Framework-derived eliminators or computation rules contribute operational payload without increasing conceptual κ.

Strict novelty computation. For molecular seeds the strict engine uses a hybrid evaluator:

$$\nu_{\text{strict}} = (\nu_G + \nu_C)_{\text{native}} + \nu_{H,\text{explicit}}.$$

The native term is computed on the conceptual evaluation telescope. The explicit topological term counts only path-style witness payload whose framework-derived computation clauses are physically present in the compiled telescope. This is the canonical source of the machine values in table 1.

Semantic minimality. Every bar-clearing candidate is subjected to semantic minimality filtering. The engine removes terminal SCC sub-bundles, re-checks the remainder for well-typedness, and rejects any candidate whose proper sub-bundle already clears the active bar. This operationalizes constructive primality in the current strict lane.

Results. The strict unguided run recovers all 15 structures in the order shown in table 1, with machine totals

$$(1, 1, 2, 5, 7, 8, 10, 18, 17, 19, 27, 35, 47, 63, 88)$$

at conceptual costs

$$(2, 1, 1, 3, 3, 3, 3, 5, 4, 4, 5, 6, 7, 9, 8).$$

The late-stage values differ from the manuscript's older hand-estimated canon; throughout this paper, the strict machine values now take precedence.

Rejected candidates. To verify that the trajectory is selective rather than merely permissive, the engine also evaluates natural but losing alternatives spanning arithmetic, classical logic, measure theory, scheme theory, elementary topos axioms, and symplectic geometry. All of them fall below the active bar at first admissibility; the comparison table is deferred to section C.

9.2 Schema Canonicality Verification

The engine still includes an exact-oracle schema mode used as a diagnostic for the early foundational steps. For steps 1–7, the depth-1 schema set using all library atoms is identical to the proof-rank schema set using only the 2-step window. This supports theorem 8.4: after schematization, depth-1 schema counting is effectively window-independent for the small foundational cases.

This diagnostic is not the manuscript's canonical novelty measure. At depth 2 the schema count explodes combinatorially, so the exact-oracle mode is now treated only as an early-step consistency check rather than as a late-stage proof of ν.

9.3 Uniform Nu Verification

Earlier drafts used a type-inhabitation comparison procedure as a uniform diagnostic. That algorithm remains useful as a historical cross-check on the foundational steps, but it is no longer the manuscript's canonical novelty source. Its late-stage counts over-amplify API and temporal shells, so the present paper does not use it to set any value in table 1.

The categorical point behind that older diagnostic remains valid: Adjoint Completion (theorem 4.22) explains why introduction data forces elimination data even when pure inhabitation counts cannot see it directly. In the current manuscript this principle is realized operationally by molecular compilation plus the strict hybrid evaluator, not by the old uniform table.

9.4 Meta-Theoretic Rule Audit

The rule audit is now an explanatory reconstruction of the strict canon rather than an independent oracle. For the foundational and geometric steps the reconstruction is exact. For the late API shells it is constrained to agree with the machine total in table 1. For DCT we record only the canonical total unless a finer split is justified directly by the current strict code path.

Results.

n	Structure	ν_{strict}	Audit status
1	Universe	1	exact foundational rule count
2	Unit	1	exact foundational rule count
3	Witness	2	exact foundational rule count
4	Π/Σ	5	exact foundational rule count
5	S^1	7	exact HIT reconstruction
6	PropTrunc	8	exact HIT reconstruction
7	S^2	10	exact HIT reconstruction
8	S^3	18	exact HIT/H-space reconstruction
9	Hopf	17	exact map reconstruction
10	Cohesion	19	exact API-shell reconstruction
11	Connections	27	expository API-shell reconstruction
12	Curvature	35	expository API-shell reconstruction
13	Metric	47	expository API-shell reconstruction
14	Hilbert	63	expository API-shell reconstruction
15	DCT	88	strict total only; no canonical finer split claimed

Remark 9.1 (Status of the DCT decomposition). The current manuscript makes a narrower claim than earlier drafts. For DCT the machine-verified total $\nu = 88$ is canonical, but the paper does not elevate any finer three-way split to theorem status unless it is exposed by the current strict evaluator itself. This keeps the manuscript aligned with the executable artifact rather than with older structural diagnostics.

Remark 9.2 (Automation scope of the verification). The verification of the PEN Synthesis Trajectory decomposes into two logically independent claims: (a) the *selection ordering and bar clearance* ($\rho_n \geq \mathrm{Bar}_n$ for all n), and (b) the *rule-component accounting* ($\nu = \nu_G + \nu_C + \nu_H$).

Claim (a) is verified computationally for **all 15 steps** by the current strict unguided run itself. Given the admissible typed candidates produced by the generation layer, the PEN quantities ν, κ, ρ, and bar clearance are computed automatically with no paper lookup and no target-conditioned evaluation term.
Claim (b)—the internal rule-component accounting—now has mixed status. For steps 1–10 the explanatory audit is exact. For steps 11–14 it is an expository reconstruction constrained to match the strict total. For step 15 only the strict total $\nu = 88$ is canonical.

The implemented finding is that the current strict unguided lane recovers the 15-step PEN Synthesis Trajectory and clears every bar mechanically. The stronger conjecture is that sufficiently powerful, materially less-pruned search over the same admissible space would recover the same trajectory without the present bounded search controls.

9.5 DCT Sensitivity Sweep

Earlier DCT sensitivity sweeps targeted the manuscript's older structural-AST calibration. Those sweeps are retained only as historical diagnostics and are not part of the current strict canon. In the present manuscript, the canonical Step-15 value is the direct strict machine total $\nu = 88$.

9.6 Cubical Agda Mechanization

The Complexity Scaling Theorem is proved in Cubical Agda [7]: for $d = 2$, $\Delta_{n+1} = \Delta_n + \Delta_{n-1}$. The proof covers initial conditions, the recurrence, the cumulative sum identity $\tau_n = F_{n+2} - 1$, and convergence $\Phi_n \to \varphi$.

Coherence Obligation Experiments trace the obligations for S^1, S^2, T^2, and the Hopf fibration in Cubical Agda. In all cases: all obligations reference at most 2 layers, at least one genuinely references 2, and none references 3.

Obligation decomposition and the abstraction barrier. Explicit obligation enumeration at steps 7–9 verifies the Integration Trace Principle (theorem 4.13): S^2 resolves $13 = 8 + 5$ obligations (8 from PropTrunc, 5 from S^1), S^3 resolves $21 = 13 + 8$ (13 from S^2, 8 from PropTrunc), and the Hopf fibration resolves $34 = 21 + 13$ (21 from S^3, 13 from S^2). The Hopf fibration (step 9) crosses the phase boundary from types to maps: the 21 domain-side obligations arise from postcomposition constraints on S^3, while the 13 codomain-side obligations arise from pullback constraints on S^2, demonstrating a natural domain/codomain duality within the two-layer window. A Cubical Agda module (`Saturation/AbstractionBarrier.agda`, `--cubical --safe`) defines L_7's interface as an opaque record and discharges all Group B obligations (8.6–8.10) from record fields alone, with no PropTrunc import. At step 9, the same methodology extends: inherited obligations for the Hopf fibration are discharged from an opaque L_8 record (`Saturation/AbstractionBarrier9.agda`). This machine-checks Sealing Encapsulation (theorem 5.10): resolved obligations are depth-1 references to the preceding layer, not depth-2 leaks to earlier layers.

9.7 Coherence Window Stress-Testing

The engine accepts a `-window d` flag parameterizing the Coherence Window. $d = 1$ (constant costs) stagnates after 4–5 structures. $d = 2$ (Fibonacci) produces the 15-structure PEN Synthesis Trajectory in the current fully coupled strict engine lane. $d = 3$ (tribonacci) stalls earlier as

costs outpace the horizon. This provides computational evidence that, among the tested values $d \in \{1, 2, 3\}$, only $d = 2$ supports sustained evolution.

10 Discussion

10.1 What Is Proved, What Is Assumed, What Is Open

Proved.

- The Coherence Window Theorems (theorems 5.7 and 5.8): $d = 1$ for extensional foundations, $d = 2$ for the fully coupled CCHM/HoTT lane studied here. The value $d = 2$ is derived from categorical first principles via three independent arguments: the Adjunction Barrier (theorem 5.14, lower bound from triangle identities), Spectral Degeneration (theorem 5.16, upper bound from obligation spectral sequence collapse at E_2), and the Clutching Family (theorem 5.18, tight lower bound from an infinite family of principal bundles).
- The Integration Trace Principle (theorem 4.13): each layer exports $|S(L_k)| = \Delta_k$ schemas, verified by explicit obligation enumeration at steps 3–9, including the phase boundary from types to maps at step 9. Machine-checked abstraction barrier at steps 8–9 in Cubical Agda. The Complexity Scaling Theorem (theorem 6.1) follows: $\Delta_n = F_n$ for $d = 2$.
- Sealing Encapsulation (theorem 5.10): resolved obligations become opaque L_k exports; machine-checked at steps 8–9 in the Cubical Agda abstraction-barrier modules.
- The Divergence of Efficiency (theorem 7.6): $\rho_n \to \infty$, strengthened by the Logarithmic Effort Growth lemma (theorem 7.5): combinatorial novelty $\Omega(n^c)$ dominates logarithmic specification cost $O(\log n)$.
- Schema Canonicality (theorem 8.4): depth-1 schemas are window-independent, verified computationally for steps 1–7.
- The $S^2 \equiv S^3$ Syntactic Identity (theorem 8.6).
- **The Topological Projection** (theorem 8.2): The formula $\nu_H = m + d^2$ is derived from the cubical path algebra within the CCHM framework. The derivation relies on framework-specific properties (connection maps, Kan operations, spectral degeneration at E_2) and should be read as a structural derivation within CCHM cubical type theory rather than a theorem of arbitrary cubical frameworks.
- The terminal DCT fixed-point signal (theorem 8.1): in the current strict unguided lane, DCT is selected at step 15 with $\nu = 88$, $\kappa = 8$, and the largest absolute overshoot in the sequence.

Assumed.

- **Fully coupled evaluation lane** (theorems 4.14 and 4.15): PEN evaluates every candidate under maximal interface density. Within that lane, full coupling is forced by canonicity preservation rather than by scoring preference; what remains assumed is that the search stays in this canonicity-preserving lane throughout.
- The Integration Trace Principle is verified for steps 3–9 and the abstraction barrier is machine-checked at steps 8–9, but a general proof for all k remains open. Steps 10–15 (modal operators and library specifications) are extrapolated: the obligation structure for axioms differs from HITs and maps, and verification of the trace principle for these steps remains open.
- The ν values in table 1: the canonical values are the machine totals produced by the current strict unguided run. Explanatory rule audits for steps 11–15 are intentionally narrower than earlier drafts and do not carry independent theorem status beyond agreement with those totals.

- The DCT novelty claim is local to the current strict evaluator: the canonical statement is $\nu = 88$ at conceptual cost $\kappa = 8$. A finer decomposition of that total remains open.

Open.

- **Specification clause boundaries for steps 1–4.** For steps 5–15, the clause count κ is unambiguous: each clause corresponds to a single inference rule, and the specification selection (theorem 4.11) is governed by the minimal-overshoot criterion. For the bootstrap steps 1–4 (Universe, Unit, Witness, Π/Σ), the clause boundaries depend on how one factors the foundational judgments. The values in table 1 follow the convention that each atomic judgment form (formation, introduction, or computation rule) is one clause; the MBTT encoding audit (section F) provides the explicit AST for verification.
- **How strong is the Step-15 claim?** The present claim is a terminal fixed-point theorem within PEN, not a universal novelty theorem over all existing mathematics. A stronger semantic novelty claim would require a broader comparative formalization program and an explicit external corpus criterion for structural equivalence classes.
- **Late-stage value decomposition.** For steps 11–15 the strict engine now gives canonical totals, but not a fully settled theorem-level decomposition into independent ν_G, ν_C, ν_H parts. The open problem is to reconcile conceptual-shell novelty and richer derived operational payload without reintroducing target-steered structural bonuses.

Conjecture versus implemented finding. The manuscript's strongest mathematical claim is a *conjecture* about search-power, not a theorem about the current runtime profile: that sufficiently powerful MBTT-first search over the same admissible space would recover the same PEN Synthesis Trajectory without materially changing the present evaluator. The *implemented finding* is narrower and fully auditable: the current codebase recovers that trajectory in a strict unguided lane using exact-band admissibility, molecular compilation, hybrid novelty evaluation, and semantic minimality filtering. The paper should therefore be read as proving PEN's objective-level theorems and reporting a strict recovery result for the current executable artifact.

10.2 Limitations

Scope of the Step-15 claim. The DCT claim is intentionally local to the PEN setup: fixed objective, fixed admissibility, fixed bounded typed search, and fixed strict selection lane. This is sufficient for a reproducible terminal-selection theorem, but not sufficient to conclude that no alternative objective or foundation could yield a different terminal structure. The result should therefore be read as a strong mechanized existence claim in one principled synthesis regime.

Objective dependence. The decisive quantities ν, κ, admissibility, and minimal overshoot are PEN's operational invariants of library growth, not canonical invariants of all possible foundational dynamics. The manuscript's theorems are therefore conditional: once these quantities are fixed by constructive irreducibility, canonicity preservation, and universal coding, the induced trajectory is deterministic.

The topological projection $\nu_H = m + d^2$ is now a derived quantity (theorem 8.2), but the derivation relies on Spectral Degeneration at E_2 and the cubical interaction matrix, which are structural properties of the path algebra rather than consequences of the PEN axioms alone. In the present manuscript that formula is used only for the explicit path-style payload preserved by the strict evaluator.

Component decomposition for axiomatic steps. For step 9 (Hopf fibration), $\nu_G = \nu_H = 0$ and $\nu = \nu_C$: as a map introducing no new type formers, all novelty is Elimination rules. For steps 10–14 (library specifications), the decomposition is dominated by ν_C (Elimination schemas

and structural interactions), with small but nonzero ν_G (2–5 Introduction schemas per step). Step 15 (DCT) is different: the current strict evaluator canonizes only the total $\nu = 88$. The full tripartite structure with all three components independently nonzero is therefore asserted in this manuscript only for the HIT steps (5, 6, 7, 8), not for the temporal shell.

The current DCT computation. The DCT's canonical Generative Capacity is the strict machine value

$$\nu(\mathrm{DCT}) = 88.$$

That value is produced by the current hybrid strict evaluator from the conceptual temporal shell plus any explicit topological witness payload preserved by compilation. The older 103 structural-AST total is no longer used as a theorem-level value in this manuscript.

10.3 Mathematical Implications

If the PEN Synthesis Trajectory faithfully identifies a locally optimal foundational trajectory under PEN, then cohesive and differential-geometric libraries arise as a consequence of efficiency optimization within the fully coupled CCHM/HoTT $d = 2$ coherence regime. The sequence exhibits a coherent arc: dependent types $\to$ spheres $\to$ fibrations $\to$ cohesion $\to$ differential geometry $\to$ temporal-cohesive synthesis with derived infinitesimal semantics.

The absorption of arithmetic is instructive: natural numbers ($\rho \approx 1.5$) and Lie groups ($\rho = 1.50$) fail the rising bar and are subsumed by more efficient geometric and modal frameworks. Within the model, efficient foundations prioritize geometric generality over discrete utility.

After DCT, PEN's external synthesis mechanism is squeezed: internal continuations are definitionally exhausted up to equivalence, while disconnected jumps are too inefficient to clear the Fibonacci-scaled bar. This suggests a computational phase transition inside the model: the search shifts from external axiomatization to internal exploration within the DCT semantic completion.

10.4 Falsifiability

The PEN framework makes three testable predictions:

1. **Dimensional limit.** If a type-theoretic formalization in this target family required Class 3 coherence (non-trivial 3-paths not reducible to 2-path coherence), the claim that $d = 2$ for the fully coupled CCHM/HoTT lane would be falsified.
2. **No early efficiency monsters.** If a structure existed with $\kappa < 4$ and $\nu > 20$ before homotopy theory, the sequence would be falsified.
3. **The DCT novelty count.** $\nu(R_{15}) = 88$ is the canonical machine value in the current strict lane. If independent analysis reduces this below the bar-clearing threshold $\kappa \cdot \mathrm{Bar}_{15} \approx 60.10$, the terminal fixed-point claim disappears and the framework is falsified.

11 Conclusion

The Principle of Efficient Novelty determines a 15-step PEN Synthesis Trajectory—15 structures constituting an emergent geometric type-theoretic framework, derived from an empty library—governed by three theorems:

1. **Coherence Windows.** In the fully coupled CCHM/HoTT intensional lane analyzed here, $d = 2$; extensional type theory has $d = 1$.
2. **Complexity Scaling.** The Integration Trace Principle—each layer's exported interface equals its integration trace—forces Fibonacci costs ($\Delta_n = F_n$) in that $d = 2$ regime; $d = 1$ stagnates.

3. **Combinatorial Novelty.** Superlinear novelty growth ensures efficiency permanently outpaces the selection bar.

and two structural results:

4. **Terminal DCT Fixed Point.** In PEN's fully coupled strict lane, the temporal-cohesive Step-15 AST is selected with $\nu = 88$, $\kappa = 8$, and maximal absolute overshoot; its semantic completion is read as DCT, and Step 16 then halts by the internal/external combinatorial squeeze. This identifies the terminal fixed point of PEN within the connected component explored by the engine.
5. **Mechanical Verification.** The strict unguided engine recovers the full 15-step sequence with the machine values reported in table 1. Explanatory rule audits and older type-inhabitation diagnostics remain secondary and do not override those totals.

The paper's final claim splits in two. *Conjecture:* with sufficiently powerful search over the same admissible MBTT space, the same PEN trajectory and same semantic identifications would emerge without the current tractability guidance. *Finding:* the present executable artifact recovers that trajectory in a strict unguided lane using exact-band admissibility, molecular compilation, hybrid novelty evaluation, and semantic minimality filtering. Taken together, these results support a conditional metatheory of typed library growth rather than an anthropomorphic discovery claim: under PEN's fixed MBTT vocabulary, admissibility policy, minimal-overshoot rule, and maximal-interface-density requirement, the fully coupled CCHM/HoTT $d = 2$ coherence regime forces Fibonacci integration debt, superlinear novelty, and convergence to a temporal-cohesive Step-15 fixed point whose semantic completion is DCT and whose external continuation is halted inside the connected component of $\mathcal{U}_0$ explored by the engine. Reproducibility details, rejected-candidate arithmetic, boundary-case clarifications, and full MBTT audits are deferred to the appendices.

A Boundary Cases and Algorithmic Limits

This appendix records four scope clarifications deferred from the main text. They address reviewer-facing boundary questions, but are logically downstream of the core derivation.

A.1 Algorithmic Criticality

The 0.11 margin at Step 6 does not indicate continuous hyperparameter tuning. In the strict MBTT lane, the relevant quantities are discrete: κ is an audited AST-clause count (theorem 4.4 and table 2) and ν_H is locked by the cubical path algebra via $\nu_H = m + d^2$ (theorem 8.2).

Theorem A.1 (Rigidity of the $d = 2$ critical window)**.** *Within the strict MBTT lane, the survival region of the PEN Synthesis Trajectory is discrete rather than continuously tunable. Hence the narrow Step-6 margin is a rigidity phenomenon, not a fitted threshold.*

Proof sketch. The sweep of theorem 8.3 ranges over integer triplets, not real-valued hyperparameters. Likewise, admissible changes to κ are discrete clause insertions or deletions in a fixed grammar. At Step 6, increasing the PropTrunc cost by one clause already moves ρ_6 from 8/3 to $2.00 < 2.56$, killing the geometric branch. The 0.11 margin is therefore the exact phase transition at which geometry first outpaces Fibonacci integration debt. □

A.2 The Algorithmic Inefficiency of the Discrete

The model does not ignore arithmetic and set-theoretic candidates; it rejects them because, in a $d = 2$ intensional regime, purely discrete structures have no homotopy bonus and only linear schema growth.

Proposition A.2 (Inefficiency of Discrete Structures)**.** *Let X be a candidate with $\nu_H(X) = 0$ and $\nu_G(X) = O(\kappa(X))$. In a $d = 2$ system, X eventually fails the selection bar.*

Proof sketch. By theorems 6.1 and 7.6, the integration bar grows with Fibonacci-governed history and diverges. If X has no higher-path contribution, its novelty remains bounded by linear growth in specification size, so $\rho(X) = \nu(X)/\kappa(X)$ stays bounded while the bar increases. Thus only candidates with nontrivial homotopy interaction can survive indefinitely. □

This is the structural reason that $\mathbb{N}$, classical logic, power-set style axioms, and ordinal arithmetic fail in table 3: they are valid mathematics, but not efficient frontier points under the PEN objective.

A.3 The Univalent Horizon

The Step-15 halt theorem is local to the connected component of $\mathcal{U}_0$ explored by the guarded internal flow. It is not a claim that mathematics globally halts.

Definition A.3 (The Univalent Horizon)**.** The *Univalent Horizon* of a library $\mathcal{B}$ is the boundary of the connected component of $\mathrm{Mod}(\mathcal{B})$ reachable from $\mathcal{B}$ by internal continuous deformation. Crossing that boundary requires an external axiomatic jump. Within PEN, the halting theorem further shows that these jumps are not merely external but also inefficient once the Step-16 interface debt is charged.

Corollary A.4 (Connected-component confinement)**.** *The DCT's guarded corecursion principle (theorem 7.9) generates structures only within the connected component of $\mathcal{U}_0$. Higher universes, large-cardinal style axioms, and non-geometric logics lie beyond that component and are unreachable by internal flow alone.*

This preserves Gödelian open-endedness. The claim is exhaustion of the largest connected fragment reachable from the empty library under PEN, not exhaustion of mathematics as such.

A.4 The DCT Novelty Count and Adjoint Completion

The DCT count $\nu = 88$ is not a raw string count. It is the canonical value produced by the current strict hybrid evaluator on the temporal shell selected at step 15. That total is the one used throughout the manuscript and in the executable selection loop.

Likewise, Adjoint Completion (theorem 4.22) is not a retrofitted scoring bonus. It is a measurement principle forced by adjoint structure in categorical logic: introduction data determines corresponding elimination structure. For bootstrap types that eliminative payload appears in ν_C; for bare HITs it is absorbed into ν_H by dimensional orthogonality, which is why the HIT rows can have $\nu_C = 0$ without contradicting adjoint completion. In the current manuscript that principle is realized through molecular compilation and strict evaluation rather than through the older uniform diagnostic.

B Artifact Map and Reproduction Checklist

The executable artifact is openly available at `https://github.com/efficient-novelty/mechanization`.

Repository layout.

- `engine/src/`: synthesis engine and scoring/selection logic.
- `engine/scripts/`: reproducibility and regression scripts.
- `agda/`: Cubical Agda mechanization artifacts.
- `pen_paper.tex`: manuscript source.

Table 3: Rejected candidates at first admissibility.

Candidate	ν	κ	ρ	Bar	Margin
Natural numbers $\mathbb{N}$	≤ 4	3	≤ 1.33	2.14 (step 5)	≤ -0.81
Classical logic (LEM)	≤ 1	1	≤ 1.00	2.14 (step 5)	≤ -1.14
Power set axiom	≤ 2	2	≤ 1.00	2.14 (step 5)	≤ -1.14
Ordinal arithmetic	≤ 6	5	≤ 1.20	2.14 (step 5)	≤ -0.94
Measure theory	≈ 12	6	≈ 2.0	4.46 (step 10)	≈ -2.5
Elementary topos axioms	≈ 15	8	≈ 1.9	4.46 (step 10)	≈ -2.6
Lie groups Lie(S^3)	9	6	1.50	4.46 (step 10)	-2.96
Scheme theory	≈ 18	7	≈ 2.6	5.47 (step 12)	≈ -2.9
Galois theory	≈ 10	5	≈ 2.0	5.47 (step 12)	≈ -3.5
Symplectic geometry	≈ 25	5	≈ 5.0	6.07 (step 13)	≈ -1.1

Minimal rerun checklist.

1. `cd engine && cabal build all`
2. `cabal run acceptance-core`
3. `cabal run ab-initio - -strict -max-steps 15 -skip-validation -skip-mcts`
4. `cabal run kolmogorov-audit` *(optional MBTT audit)*

These commands are sufficient to inspect the implementation, rerun strict synthesis, execute the core acceptance harness, and regenerate the MBTT complexity audit when needed.

C Worked Arithmetic and Rejected Candidates

C.1 Worked arithmetic for Steps 5–6

The transition from the Circle to Propositional Truncation is the tightest part of the trajectory and is therefore the most useful arithmetic checkpoint. At Step 5, the Circle has $\Delta_5 = F_5 = 5$, $\nu_5 = 7$, $\kappa_5 = 3$, and $\rho_5 = 7/3 = 2.33$. The structural inflation is $\Phi_5 = \Delta_5/\Delta_4 = 5/3 = 1.67$, the cumulative baseline is $\Omega_4 = (1 + 1 + 2 + 5)/(2 + 1 + 1 + 3) = 9/7 = 1.29$, and the bar is $\mathrm{Bar}_5 = \Phi_5\Omega_4 = 2.14$. Thus S^1 survives with margin 0.19.

At Step 6, Propositional Truncation has $\Delta_6 = F_6 = 8$, $\nu_6 = 8$, $\kappa_6 = 3$, and $\rho_6 = 8/3 = 2.67$. Now $\Phi_6 = 8/5 = 1.60$, $\Omega_5 = (1 + 1 + 2 + 5 + 7)/(2 + 1 + 1 + 3 + 3) = 16/10 = 1.60$, and $\mathrm{Bar}_6 = 2.56$. Hence Step 6 survives by only 0.11. If PropTrunc required one extra clause, its efficiency would fall to 2.00 and the geometric branch would terminate immediately.

C.2 Rejected candidates

The candidates in table 3 are evaluated at their first admissible step. They are included to show that PEN does not merely favor one hand-picked branch; it scores and rejects multiple mathematically natural alternatives under the same objective.

Discrete candidates. $\mathbb{N}$, classical logic, power-set style axioms, and ordinal arithmetic are all trapped by the same mechanism: they remain 0-groupoid or near-0-groupoid structures, so $\nu_H = 0$ and novelty grows at most linearly in specification size. For $\mathbb{N}$ in particular, addition and multiplication do not increase κ-worthy novelty because they are definitionally derivable from $\mathsf{rec}_{\mathbb{N}}$.

Generous arithmetic bundling. Even an intentionally over-generous arithmetic package with metric-scale clause count $\kappa = 7$ still lacks the homotopy term. Its best-case efficiency remains subcritical once the bar reaches the geometric regime, so geometry wins by topology-driven interaction growth rather than by curated bundling.

Higher but losing geometric or algebraic packages. Measure theory, elementary toposes, scheme theory, and Galois theory generate genuine structure but remain too discrete in the present scoring regime. Symplectic geometry comes closest to survival, but by Step 13 it still falls below the bar cleared by metric structure.

D DCT: Key Theorems and Derived Construction Families

D.1 Internal Tangent Bundle

Theorem D.1. *For any type $X : \mathcal{U}$ in DCT, the tangent bundle is: $TX := \sum_{x:X} (X^{\mathbb{D}})_x$, where $(X^{\mathbb{D}})_x$ is the type of infinitesimal curves through x. The projection $\pi : TX \to X$ is a fiber bundle, and any flow on X lifts to TX.*

D.2 Temporal Evolution

Theorem D.2. *Any type $X : \mathcal{U}$ in DCT admits a temporal evolution operator $E_X : \mathbb{R} \to (X \to X)$ satisfying identity, group property, smoothness, and discrete-structure preservation. The next-modality $\bigcirc X$ is the type of fixed points of infinitesimal evolution.*

D.3 Symplectic Flow Corollary

Theorem D.3. *For a smooth type M with symplectic form ω: any $H : M \to \mathbb{R}$ generates a unique symplectic vector field X_H; the resulting flow preserves ω; the Poisson bracket makes $C^\infty(M)$ a Lie algebra.*

Corollary D.4 (PDE-style evolution as internal flow)**.** *Any linear PDE $\partial_t u = \mathcal{L}u$ on a smooth type is an instance of temporal evolution:*

$$E(t) = e^{t\mathcal{L}}.$$

Heat, wave, and Schrödinger-type equations are distinguished by algebraic properties of $\mathcal{L}$.

D.4 Representative Formalization Families

Temporal logic formulas, guarded recursion, geometric flow schemata, and symplectic transport schemata are all instances of DCT's temporal evolution operator. Representative families include:

- **Classical mechanics:** on T^*Q, Hamiltonians generate flows obeying Hamilton's equations via theorem D.3.
- **Gauge dynamics:** time-sliced Yang–Mills evolution is represented by flow on connection data coupled to cohesive transport structure.
- **Geometric flows:** Ricci/mean-curvature-style equations are instances of vector-field-generated evolution on moduli spaces.
- **Temporal verification:** LTL constructors $(\bigcirc, \Diamond)$ and guarded recursion internalize safety/liveness-style reasoning.

D.5 Relation to Existing Frameworks

The ingredients of DCT each have established antecedents:

- Cohesive toposes provide $\flat, \sharp, \Pi, \mathrm{Disc}$ but no native temporal modality [4, 3].
- Guarded temporal type theory provides the "next" modality for recursion but not the cohesive semantic transfer studied here [5].
- Synthetic differential geometry provides infinitesimals and tangent reasoning but not the temporal-cohesive exchange laws selected by the Step-15 AST [22].

What is specific to DCT in this manuscript is the unified typed synthesis of a temporal-cohesive AST together with the proof that its semantic completion carries internal tangent, temporal, and symplectic-flow consequences.

D.6 Semantic Audit of DCT Novelty

The current manuscript canon is the strict machine value

$$\nu(\mathrm{DCT}) = 88.$$

This value is produced by the current strict hybrid evaluator and is the only DCT novelty total used in the main text. The earlier structural-AST total of 103 is retained only as a historical diagnostic and is not treated as a theorem-level quantity here.

What the current code supports. The selected temporal shell still justifies the same qualitative construction families:

- temporal endomorphism and guarded recursion patterns built from $\bigcirc$ and $\Diamond$,
- spatial–temporal exchange laws linking temporal operators to cohesive modalities,
- transport and flow-like interaction with the already selected cohesive and differential layers.

What changes is the measurement claim. The paper now records only the strict machine total and avoids promoting a finer semantic bucketization that is not exposed by the current evaluator.

Representative schema examples. The following shapes remain representative of the mathematical content carried by the temporal shell:

1. **Temporal endomorphism flow:** $\Pi_{A:\mathcal{U}}\ \bigcirc(\flat A \to A)$, schematized as $\bigcirc(L \to L)$.
2. **Connection transport shape:** $\Pi_{A:\mathcal{U}}\ \sharp(\nabla(A)) \to \bigcirc(A)$, schematized as $\sharp(L) \to \bigcirc(L)$.
3. **Guarded recursion / step-indexing:** $\Pi_{A:\mathcal{U}}\ \bigcirc(A) \to A$, schematized as $\bigcirc(L) \to L$.

These examples motivate the DCT interpretation, but the canonical novelty claim is the strict total 88, not a larger post hoc semantic grouping.

E Semantic Identification of Selected ASTs (Steps 10–14)

When the synthesis engine selects the next bar-clearing candidate under the minimal-overshoot rule, the raw representation is an anonymous MBTT AST (section F). The algorithm itself does not assign semantic labels such as "metric" or "Hilbert"; those names are post-hoc identifications. This appendix documents that translation layer for Steps 10–14 by giving the corresponding formal type signatures and clarifying their axiomatic status.

The PEN Synthesis Trajectory's first nine steps—Universe through Hopf fibration—are standard constructions in HoTT, formalizable directly in Cubical Agda. Steps 10–14 extend the type theory with differential-geometric structure. This appendix provides the exact formal type signatures for each, clarifying their axiomatic status and showing how the inference-rule counts (ν_G, ν_C) arise.

Axiomatic status. Steps 10–14 are *library specifications* (theorem 4.19): packages of types, operations, and axioms that expand the library's derivation repertoire. Unlike defined terms (which have $\nu = 0$ because they name existing derivable judgments), these specifications introduce structure that is *not derivable* from the prior library. Each step adds Introduction schemas (unit maps, constructors, type-formation side-effects) and Elimination schemas (counits, structural operations, cross-interactions). The ν counts reflect the total number of derivation schemas natively unlocked by the specification, scored by the same methodology used for foundational steps 1–9.

E.1 Step 10: Cohesion (Adjoint Quadruple of Modalities)

Following Lawvere [4] and Schreiber [3], the cohesive structure extends HoTT with an adjoint string of modalities:

$$\Pi_\infty \dashv \flat \dashv \sharp$$

where Π_∞ is shape (fundamental ∞-groupoid), $\flat$ is flat (discrete), and $\sharp$ is sharp (codiscrete).

Type formation axioms ($\kappa = 4$ clauses).

$\flat$-form:	$A : \mathcal{U} \vdash \flat A : \mathcal{U}$
$\sharp$-form:	$A : \mathcal{U} \vdash \sharp A : \mathcal{U}$
Disc-form:	$A : \mathcal{U} \vdash \mathrm{Disc}(A) : \mathcal{U}$
Π_{coh}-form:	$A : \mathcal{U} \vdash \Pi_\infty(A) : \mathcal{U}$

Derived inference rules ($\nu = 19$: $\nu_G = 2$, $\nu_C = 17$). The adjoint structure generates three families of rules:

(i) Unit/counit maps (4 rules: 2 Introduction, 2 Elimination):

$\flat$-counit:	$\flat A \to A$	(ν_C)
$\sharp$-unit:	$A \to \sharp A$	(ν_G)
Π_∞-unit:	$A \to \Pi_\infty(A)$	(ν_G)
Disc-counit:	$\mathrm{Disc}(\Gamma(A)) \to A$	(ν_C)

The unit maps ($\eta_\sharp$, η_{Π_∞}) are Introduction rules: they construct new term inhabitants. The counit maps ($\varepsilon_\flat$, $\varepsilon_{\mathrm{Disc}}$) are Elimination rules: they extract underlying data from a modal type.

(ii) Modal structural rules (9 rules, all ν_C): Idempotency ($\flat\flat A \simeq \flat A$, $\sharp\sharp A \simeq \sharp A$, $\Pi_\infty\Pi_\infty A \simeq \Pi_\infty A$); distribution over Π and Σ types ($\flat(\Pi_{(a:A)}B) \to \Pi_{(a:\flat A)}\flat B$, and analogously for $\sharp$, Π_∞); these yield 3 idempotency + 6 distribution = 9 structural rules.

(iii) Cross-interactions with HITs (6 rules, all ν_C): Each modality interacts with the library's higher inductive types (S^1, S^2, S^3, Hopf): $\flat(S^1) \simeq \mathbf{1}$ (the discrete circle is contractible), $\Pi_\infty(S^1) \simeq B\mathbb{Z}$ (the shape of the circle classifies $\mathbb{Z}$-torsors), and analogous interactions for the other HITs. These produce 6 rules from 3 modalities $\times$ 2 independent HIT interactions.

Total: $2\ (\nu_G) + 2 + 9 + 6\ (\nu_C) = 19$.

E.2 Step 11: Connections (∇ on Cohesive Types)

A connection provides the differential-geometric machinery for parallel transport. The formal content is a splitting of the de Rham coefficient sequence in the cohesive setting.

Axioms ($\kappa = 5$ clauses).

∇-form:	$\nabla : \flat(TX) \to TX$	(connection operator)
transport$_\nabla$:	$\mathrm{Id}_X(x, y) \to (F_x \simeq F_y)$	(parallel transport)
cov$_\nabla$:	$\flat(A) \to A$	(covariant derivative)
lift$_\nabla$:	$\flat(TX) \to TX$	(horizontal lift)
Leibniz:	$\nabla(f \cdot s) = df \otimes s + f \cdot \nabla s$	(product rule)

Derived inference rules ($\nu = 27$: $\nu_G = 2$, $\nu_C = 25$). Of the 5 self-interaction rules, 2 are Introduction-type (transport and horizontal lift each construct new terms), while 3 are Elimination-type. The remaining rules are Elimination schemas: 6 cross-interactions with cohesive modalities (each of ∇, transport, cov interacts with $\flat$ and $\sharp$) + 2 function-space rules (∇ on Π- and Σ-types) + 14 library cross-interactions (connections on S^1, S^2, S^3, Hopf bundles, and modality compositions). Total: $2\ (\nu_G) + 3 + 6 + 2 + 14\ (\nu_C) = 27$.

Why not definable. A connection on a general cohesive type X is not derivable from the modalities $\flat$, $\sharp$ alone. The de Rham coefficient sequence $\flat X \to X \to \Pi_{\infty\mathrm{dR}} X$ is exact, but a splitting of this sequence (i.e., a connection) is additional data requiring an axiom.

E.3 Step 12: Curvature ($R = d\nabla + \nabla \wedge \nabla$)

The curvature measures the failure of a connection to be flat.

Axioms ($\kappa = 6$ clauses).

R-form:	$R : \nabla \to \Omega^2(X)$	(curvature 2-form)
Bianchi:	$dR + [\nabla, R] = 0$	(Bianchi identity)
hol:	$\Omega(X, x) \to \mathrm{Aut}(F_x)$	(holonomy)
CW:	$R \to H^*(X)$	(Chern–Weil map)
char:	$\Omega^*(BG) \to H^*(X)$	(characteristic classes)
R-comp:	$\nabla_1, \nabla_2 \mapsto R(\nabla_1 \circ \nabla_2)$	(composition)

Derived inference rules ($\nu = 35$: $\nu_G = 2$, $\nu_C = 33$). Of the 6 self-interaction rules, 2 are Introduction-type (the curvature 2-form R and holonomy map each construct new mathematical objects), while 4 are Elimination-type (Bianchi identity, Chern–Weil homomorphism, characteristic classes, composition). The remaining rules are Elimination schemas: 8 cross-interactions with cohesion and connections + 2 function-space rules + 19 library cross-interactions. Total: $2\ (\nu_G) + 4 + 8 + 2 + 19\ (\nu_C) = 35$.

Why not definable. While the curvature formula $R = d\nabla + \nabla \wedge \nabla$ is algebraically determined by a connection, the *elimination rules* it generates (Bianchi identity, holonomy, Chern–Weil theory) require the curvature as a first-class object with its own structural rules. These rules expand the derivation logic in ways not accessible from the connection alone.

E.4 Step 13: Metric ($g : TX \otimes TX \to \mathbb{R}$)

A Riemannian metric is a symmetric, positive-definite bilinear form on the tangent bundle.

How $\mathbb{R}$ is typed (explicit dependency accounting). The metric codomain $\mathbb{R}$ is not available by default. We evaluate explicit scalar routes under strict first-use charging: Cauchy-style arithmetic completion, topological arithmetic compression, and synthetic continuum insertion. For a route with scalar burden κ_{scalar}, Step 13 is scored by

$$\rho_{13}^{\mathrm{strict}} = \frac{47 + \Delta\nu_{\mathrm{mech}}}{7 + \kappa_{\mathrm{scalar}}}, \qquad \Delta\nu_{\min} = \lceil 6.07\,(7 + \kappa_{\mathrm{scalar}}) - 47 \rceil\,.$$

Resolution of the Ring Structure of the Scalar Field. A natural objection to extracting the metric codomain $\mathbb{R}$ from the cohesive circle $\Sigma_{x:S^1}(\eta(x) = \mathsf{base})$ is that this yields $\mathbb{R}$ only as an additive group (via path concatenation), whereas a metric requires $\mathbb{R}$ to be a commutative ring. As the algorithmic experiments show (`full_postulated_commutative_ring`), externally axiomatizing this ring structure can add up to 15 clauses, dropping efficiency to $\rho \approx 3.22$ and failing the Step 13 survival bar. This is exactly the same discrete-inefficiency mechanism in action: the $d = 2$ universe cannot afford discrete algebraic axiomatization.

However, the PEN algorithm natively avoids this integration debt. In cohesive HoTT, the full commutative ring structure of $\mathbb{R}$ is constructively derivable. Defining the scalar field as the space of continuous group endomorphisms of the real line,

$$\mathcal{R} := \mathrm{End}_{\mathrm{Grp}}(\mathbb{R}_{\mathrm{add}}),$$

multiplication is definitionally equivalent to function composition $(f \circ g)$. The core ring axioms (associativity, distributivity, multiplicative identity) are therefore not external postulates requiring additional κ-cost, but definitional tautologies of dependent function type structure (Π). By the Constructive Irreducibility Boundary (Axiom 3.5), the commutative ring structure is algorithmically free ($\kappa_{\mathrm{scalar}} = 0$). The framework thus shows that continuous algebra is not axiomatized as separate discrete baggage; it is inherited natively from topology and type theory.

Axioms ($\kappa = 7$ clauses).

g-form:	$g : TX \otimes_s TX \to \mathbb{R}$	(symmetric bilinear form)
LC(g):	$g \mapsto \nabla_g$	(Levi-Civita connection)
γ-form:	$[0,1] \to X$	(geodesic equation)
vol_g:	$g \mapsto \omega \in \Omega^n(X)$	(volume form)
$\star_g$:	$\Omega^k(X) \to \Omega^{n-k}(X)$	(Hodge star)
Δ_g:	$\Omega^*(X) \to \Omega^*(X)$	(Laplace–Beltrami operator)
Ric_g:	$g \mapsto (\mathrm{Ric}, R_{\mathrm{scalar}}) \in \Omega^2(X) \times \mathbb{R}$	(Ricci tensor / scalar curvature)

Derivation schemas ($\nu = 47$: $\nu_G = 4$, $\nu_C = 43$). Of the 7 self-interaction schemas, 4 are Introduction-type (the metric g, Levi-Civita connection ∇_g, geodesic equation, and volume form each construct new objects), while 3 are Elimination-type (Hodge star, Laplacian, Ricci extraction). The remaining schemas are Elimination-type: the structural interaction count $\nu_C = \max_{\mathrm{ref}} \nu + \kappa + (\text{distinct refs} - 1) = 35 + 7 + 1 = 43$, capturing the bounded coupling of the metric with its directly referenced library entries (Curvature at $\nu = 35$, Connections). Total: $4\ (\nu_G) + 43\ (\nu_C) = 47$. These schemas are counted as *derivation patterns natively unlocked* by the metric specification, not as inference rules of the core type theory (theorem 4.19). The Hodge star, for example, becomes available only after the metric is specified; it is not derivable from prior structures.

Why not definable. In cohesive HoTT, types carry smooth structure via the adjoint modalities, but no intrinsic notion of distance or angle. A metric is additional structure: the Levi-Civita theorem (uniqueness of the torsion-free metric connection) guarantees that each metric determines a unique connection, but the metric itself is not derivable from cohesion.

Derivability of components vs. $\mathbb{N}$'s addition. A natural objection asks why the Levi-Civita connection, Hodge star, Laplacian, and Ricci scalar are counted as specification clauses rather than derivable consequences with $\nu = 0$—the standard applied to $\mathbb{N}$'s addition from $\mathrm{rec}_{\mathbb{N}}$. The distinction is *constructive availability within the type theory.* Addition on $\mathbb{N}$ is a definitional

term: add $:= \mathrm{rec}_{\mathbb{N}}\,(\lambda n.\,\mathrm{succ}\,n)$ is a closed expression in MLTT requiring no additional axioms. By contrast, the Levi-Civita connection is the solution to a system of partial differential equations (torsion-freeness + metric compatibility) whose existence and uniqueness require analytic machinery—the Fundamental Theorem of Riemannian Geometry—that is not constructively available from the bare metric tensor g within cohesive HoTT. The Hodge star requires extending the inner product induced by g from TX to the full exterior algebra $\Lambda^k T^*X$, a construction that depends on the orientation and dimension of X; the Laplacian and Ricci scalar each require the connection as input. These operations form a *dependency chain*—$g \to \nabla_g \to R \to \mathrm{Ric}$—in which each link requires existence theorems that are not reducible to a single recursive definition. The specification bundles this chain as a coherent API package (theorem 4.30, structural unity): the components are interdependent and cannot be meaningfully decomposed. A stripped-down specification with $\kappa = 1$ (bare g-tensor alone) would have $\nu_G = 1$, $\nu_C \leq 2$, giving $\rho \leq 3.0$—below the Step 13 bar of 6.07. The metric clears the bar precisely because its API specification natively unlocks the full chain of derived geometric operations as new derivation schemas.

E.5 Step 14: Hilbert Functional (Operator Algebra)

The Hilbert functional introduces inner-product structure, completeness, and spectral theory—a high-interaction analytic package over the geometric library.

Axioms ($\kappa = 9$ clauses).

$\langle\cdot,\cdot\rangle$-form:	$V \times V \to \mathbb{R}$	(inner product)
complete:	$\mathrm{Cauchy}(V) \to V$	(Cauchy completeness)
$\perp$-decomp:	$V \to V_1 \oplus V_2$	(orthogonal decomposition)
spec:	$\mathrm{Op}(V) \to \Sigma(\lambda : \mathbb{R}).\, V_\lambda$	(spectral decomposition)
C^*-alg:	$(V \to V) \times (V \to V) \to (V \to V)$	(C^*-algebra structure)
g-compat:	$\mathrm{Metric} \to \langle\cdot,\cdot\rangle$	(metric compatibility)
R-op:	$\mathrm{Curv} \to \mathrm{Op}(V)$	(curvature operator)
∇-op:	$\mathrm{Conn} \to \mathrm{Op}(V)$	(connection operator)
$\delta/\delta\phi$:	$(V \to \mathbb{R}) \to V$	(functional derivative)

Derived inference rules ($\nu = 63$: $\nu_G = 5$, $\nu_C = 58$). Of the spectral-theoretic rules, 5 are Introduction-type (inner product, completeness/limits, orthogonal decomposition, spectral decomposition, and the functional derivative each construct new objects). The remaining 58 rules are Elimination schemas: the structural interaction count $\nu_C = \max_{\mathrm{ref}} \nu + \kappa + (\text{distinct refs} - 1) = 47 + 9 + 2 = 58$, capturing the bounded coupling of Hilbert structure with its directly referenced library entries (Metric at $\nu = 47$, Curvature, Connections). Total: $5\ (\nu_G) + 58\ (\nu_C) = 63$.

Why not definable. Hilbert-space structure—inner products, completeness, spectral theory—requires analytic axioms beyond the algebraic and geometric content of steps 10–13. The functional derivative $\delta/\delta\phi$ provides variational-calculus machinery that couples operator structure to the metric/connection/curvature layer.

Selection margin at Step 14. The Hilbert functional clears the selection bar by 0.22 (table 1): $\rho = 63/9 = 7.00$ against Bar $= 6.78$. While not the tightest margin (that distinction belongs to Propositional Truncation at step 6 with margin 0.11), it remains structurally significant: the analytic content of Hilbert-space theory ($\nu = 63$) survives only because its 9 specification clauses achieve sufficient efficiency. If the analytic axioms required even two additional

Table 4: Complete encoding audit. κ is the clause count (used in the main text); MBTT is the total bit-length of all clauses in the prefix-free encoding. The specification shown for each step is the one *selected* by the minimal-overshoot criterion (theorem 4.33).

n	κ	Structure	Clauses	MBTT (bits)	Selected specification
1	2	Universe	1	4	$\mathcal{U}$-formation
2	1	Unit	1	10	$\mathbf{1}$-formation
3	1	Witness	2	27	$\star$ + ind_1
4	3	Π/Σ	5	93	λ, pair, app, fst, snd
5	3	S^1	3	21	base + loop
6	3	PropTrunc	3	35	$\lVert\cdot\rVert_0$ + squash
7	3	S^2	3	23	base + surf†
8	5	S^3	5	66	H-space: base, 3-cell, μ, unit†
9	4	Hopf	4	78	$h : S^3 \to S^2$ + fiber + section
10	4	Cohesion	4	45	$\flat, \sharp, \mathrm{Disc}, \Pi_{\mathrm{coh}}$
11	5	Connections	5	79	∇, transport, covariant lift, Leibniz
12	6	Curvature	6	121	R, Bianchi, holonomy, Chern–Weil, char. class
13	7	Metric	7	138	See table 5
14	9	Hilbert	9	169	See table 6
15	8	DCT	8	229	See table 7

specification clauses ($\kappa = 11$, giving $\rho \approx 5.73 < 6.78$), the sequence would permanently stagnate at Riemannian geometry without reaching this operator-analytic layer.

F The MBTT Encoding Audit

This appendix opens the κ "black box" by reporting the encoding-specific audit data for all 15 genesis steps. The shared MBTT grammar is fixed in table 2; here we restrict attention to prefix-cost conventions, Elias-gamma pointer costs, and the full per-step bit-length audit. The prefixes in table 2 satisfy the Kraft–McMillan inequality by construction.

Elias Gamma coding. For a positive integer n, $|\gamma(n)| = 2\lfloor \log_2 n \rfloor + 1$ bits. Examples: $\gamma(1) = 1$ bit; $\gamma(2)$–$\gamma(3) = 3$ bits; $\gamma(4)$–$\gamma(7) = 5$ bits; $\gamma(8)$–$\gamma(15) = 7$ bits. Library pointers use 1-based indexing: $\textsc{Lib}(1)$ references the first library entry (Universe) and costs $3 + 1 = 4$ bits; $\textsc{Lib}(12)$ references Curvature and costs $3 + 7 = 10$ bits.

F.1 Full Audit: Clause Counts vs. MBTT Bit-Lengths

†For S^2 and S^3, shorter suspension encodings exist (ΣS^1 and ΣS^2, each 13 bits / 1 clause), but the minimal-overshoot criterion (theorem 4.33) selects the native or enriched specifications because they yield tighter efficiency matches to the bar (theorem 4.11).

The MBTT bit-lengths range from 4 to 229 across the PEN Synthesis Trajectory, while clause counts range from 1 to 9. The two measures are positively correlated (Spearman $r \approx 0.87$) but not identical: steps that reference deep library entries (like DCT at 229 bits) have disproportionately large bit-lengths due to Elias gamma costs. The main text uses clause counts for κ because they are encoding-independent; the bit-lengths serve as an independent verification layer.

F.2 Per-Clause Breakdown: Steps 13–15

We now provide the explicit AST and bit-cost for the three steps with the tightest selection margins.

Table 5: Step 13: Metric + frame bundle ($\kappa = 7$ clauses, 138 bits total). Library references: Connections at index 11 ($\textsc{Lib}(11) = 10$ bits), Curvature at index 12 ($\textsc{Lib}(12) = 10$ bits).

#	Clause	AST	Bits
1	Symmetric bilinear form g	$\Sigma(\Pi(\mathrm{Var}_1, \mathrm{Var}_1), \Pi(\mathrm{Var}_1, \mathrm{Var}_1))$	26
2	Levi-Civita connection	$\Pi(\Sigma(\mathrm{Var}_1, \mathrm{Var}_2), \textsc{Lib}(11))$	27
3	Geodesic equation	$\Pi(\mathrm{Var}_1, \Pi(\mathrm{Var}_1, \mathrm{Var}_1))$	18
4	Volume form	$\mathrm{Lam}(\mathrm{App}(\mathrm{Var}_1, \mathrm{Var}_2))$	14
5	Hodge star	$\Pi(\textsc{Lib}(12), \textsc{Lib}(12))$	23
6	Laplacian	$\mathrm{Lam}(\Pi(\mathrm{Var}_1, \mathrm{Var}_1))$	13
7	Ricci scalar	$\Pi(\textsc{Lib}(12), \mathrm{Var}_1)$	17
	Total		**138**

Table 6: Step 14: Hilbert functional ($\kappa = 9$ clauses, 169 bits total). Library references: Connections at index 11, Curvature at index 12, Metric at index 13 ($\textsc{Lib}(13) = 10$ bits).

#	Clause	AST	Bits
1	Inner product	$\Sigma(\Pi(\mathrm{Var}_1, \Pi(\mathrm{Var}_1, \mathcal{U})), \mathrm{Var}_1)$	26
2	Completeness	$\Pi(\mathrm{Var}_1, \mathrm{Var}_1)$	11
3	Orthogonal decomposition	$\Pi(\mathrm{Var}_1, \Sigma(\mathrm{Var}_1, \mathrm{Var}_1))$	19
4	Spectral decomposition	$\Pi(\mathrm{Lam}(\mathrm{Var}_1), \Sigma(\mathrm{Var}_1, \mathrm{Var}_2))$	23
5	C^*-algebra structure	$\Sigma(\Pi(\mathrm{Var}_1, \mathrm{Var}_1), \Pi(\mathrm{Var}_1, \mathrm{Var}_1))$	26
6	Metric compatibility	$\Pi(\textsc{Lib}(13), \mathrm{Var}_1)$	17
7	Curvature operator	$\Pi(\textsc{Lib}(12), \mathrm{Var}_1)$	17
8	Connection lift	$\Pi(\textsc{Lib}(11), \mathrm{Var}_1)$	17
9	Functional derivative	$\mathrm{Lam}(\Pi(\mathrm{Var}_1, \mathcal{U}))$	13
	Total		**169**

Table 7: Step 15: temporal-cohesive extension (semantic DCT completion) ($\kappa = 8$ clauses, 229 bits total). Library reference: Cohesion at index 10 ($\textsc{Lib}(10) = 10$ bits). The audited AST contains temporal operators and exchange clauses only; the infinitesimal reading is derived semantically rather than encoded as a primitive node.

#	Clause	AST	Bits
1	Next modality $\bigcirc X$	$\bigcirc(\mathrm{Var}_1)$	13
2	Eventually modality $\Diamond X$	$\Diamond(\mathrm{Var}_1)$	13
3	$\bigcirc \to \Diamond$ axiom	$\Pi(\bigcirc(\mathrm{Var}_1), \Diamond(\mathrm{Var}_1))$	29
4	Spatial–temporal interaction	$\mathrm{Lam}(\mathrm{App}(\textsc{Lib}(10), \bigcirc(\mathrm{Var}_1)))$	27
5	$\flat\bigcirc \leftrightarrow \bigcirc\flat$ exchange	$\Pi(\flat(\bigcirc(\mathrm{Var}_1)), \bigcirc(\flat(\mathrm{Var}_1)))$	43
6	$\sharp\Diamond \leftrightarrow \Diamond\sharp$ exchange	$\Pi(\sharp(\Diamond(\mathrm{Var}_1)), \Diamond(\sharp(\mathrm{Var}_1)))$	43
7	$\Diamond$-elimination	$\mathrm{Lam}(\mathrm{App}(\Diamond(\mathrm{Var}_1), \mathrm{Var}_2))$	23
8	$\bigcirc\bigcirc \to \bigcirc$ witness	$\Pi(\bigcirc(\bigcirc(\mathrm{Var}_1)), \bigcirc(\mathrm{Var}_1))$	38
	Total		**229**

F.3 Canonicality of the Encoding

The MBTT grammar is not gerrymandered to produce the PEN Synthesis Trajectory. Three properties establish this:

1. **Standard prefix-free design.** The prefix lengths follow the standard entropy-coding principle: constructors that appear most frequently in dependent type theory (App, Lam, Pi) receive the shortest prefixes (2–3 bits), while domain-specific operators (modal, temporal) receive longer ones (7–9 bits). This is the same design principle used in Huffman and arithmetic codes. Crucially, the modal/temporal nodes are not "rewarded" by this encoding; they are assigned *higher* local description cost than core constructors.
2. **Separation of concerns.** The main text uses κ = clause count, which is entirely encoding-independent. The MBTT bit-lengths serve only as an independent audit: they verify that no clause is degenerate (encoding a trivial operation in a complex AST) and that the total specification complexity is monotonically increasing across the PEN Synthesis Trajectory.
3. **Reproducibility.** The complete MBTT encoding is implemented in `Kolmogorov.hs` (under `engine/src/`). Running `cabal run kolmogorov-audit` in the engine directory produces table 4 deterministically. Because all candidate families are scored in this single grammar, the result is an apples-to-apples optimum test over combinations, not a hand-picked proof that one pre-named structure must win.

F.4 Search-Space Tractability and Strict Recovery

A recurring objection is that the MBTT audit lengths (e.g., 138 bits for Metric, 229 bits for DCT) imply an infeasible raw space of 2^{138} or 2^{229} binary strings. That objection would be valid for blind bitstring search; it does not describe the implemented engine.

The runtime search is constrained by construction:

1. **Typed manifold, not raw strings.** Candidates are generated as well-formed MBTT telescopes or molecular signatures, not as unconstrained binaries.
2. **Exact-band control.** Enumeration runs under strict admissibility caps and prioritized exact κ-bands rather than over the full raw grammar at once.
3. **Molecular-first generation.** Ambient formers, HIT signatures, and API shells are generated at the signature level and only then compiled to MBTT payload.
4. **Structural pruning.** Ill-typed and inadmissible telescopes are removed by type checks, derivability checks, and semantic minimality before final selection.
5. **Conceptual/compiled separation.** Conceptual κ is charged on the signature while the compiled payload is evaluated for novelty, sharply shrinking the search burden for later steps.
6. **Atomic fallback only when needed.** Raw MBTT enumeration remains available, but only as a fallback when no molecular candidate inhabits the active band.
7. **Diagnostics separated from canon.** Reference telescopes and older diagnostic lanes may still be evaluated for validation, but they do not set the manuscript's canonical values.

Hence the implementation is a strict unguided recovery lane over an admissible, bounded, and reproducible candidate manifold—not exhaustive brute-force over the entire binary code space. The MBTT bit-lengths in table 4 are post-hoc representation audits of selected specifications, not the dimensionality of the runtime enumeration domain.

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